

MR Prerequisite: Storage and Collection of Recyclables

This prerequisite applies to

- ▶ BD+C: New Construction
- ▶ BD+C: Core & Shell
- ▶ BD+C: Schools
- ▶ BD+C: Retail
- ▶ BD+C: Data Centers
- ▶ BD+C: Warehouses & Distribution Centers
- ▶ BD+C: Hospitality
- ▶ BD+C: Healthcare

INTENT

To reduce the disproportionate burden of landfills and incinerators that is generated by building occupants' waste hauled to and disposed of in landfills and incinerators through reduction, reuse and recycling service and education, and to conserve natural resources for future generations.

REQUIREMENTS

NC, CS, SCHOOLS, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY NC, HEALTHCARE, RETAIL

Provide dedicated areas accessible to waste haulers and building occupants for the collection and storage of recyclable materials for the entire building. Collection and storage areas may be separate locations. Recyclable materials must include mixed paper, corrugated cardboard, glass, plastics, and metals. Take appropriate measures for the safe collection, storage, and disposal of two of the following: batteries, mercury-containing lamps, and electronic waste.

GUIDANCE

There are no substantive changes to the prerequisite requirements; refer to the LEED v4 reference guide

Beta Update

Retail projects should now follow guidance for New Construction. Retail-specific requirements are no longer applicable.

Further Explanation

Connection to Ongoing Performance

- ▶ LEED O+M MR prerequisite Waste Performance: This prerequisite is a strategy that can help achieve the MR prerequisite: Waste Performance in the v4.1 O+M rating system.

MR Prerequisite: PBT Source Reduction – Mercury

This prerequisite applies to

- ▶ BD+C: Healthcare

INTENT

To reduce mercury-containing products and devices and mercury release through product substitution, capture, and recycling.

REQUIREMENTS

HEALTHCARE

As part of the project's recycling collection system, identify the following:

- ▶ types of mercury-containing products and devices to be collected;
- ▶ criteria governing how they are to be handled by a recycling program; and
- ▶ disposal methods for captured mercury.

Applicable mercury-containing products and devices include, but are not limited to, lamps (such as linear and circular fluorescents, integrally ballasted and nonintegrally ballasted compact fluorescents and HIDs) and dental wastes (such as scrap amalgam, chair side traps, and separator wastes).

In facilities delivering dental care, specify and install amalgam separation devices that meet or exceed the ISO-11143 standard.

Comply with the mercury elimination requirements outlined below, from the 2010 FGI Guidelines for Design and Construction of Health Care Facilities, Section A1.3- 4b, Mercury Elimination.

- ▶ 4.2.1.1. New construction: healthcare facilities may not use mercury-containing equipment, including thermostats, switching devices, and other building system sources. Lamps are excluded.
- ▶ 4.2.1.2. Renovation: healthcare facilities must develop a plan to phase out mercury-containing products and upgrade current mercury-containing lamps to high-efficiency, low-mercury, or mercury-free lamp technology.

Do not specify or install preheat, T-9, T-10, or T-12 fluorescents or mercury vapor high-intensity discharge (HID) lamps in the project. Do not specify probe-start metal halide HID lamps in any interior spaces.

Specify and install illuminated exit signs that do not contain mercury and use less than 5 watts of electricity.

Fluorescent and high-pressure sodium lamps must meet the criteria in Table 1.

Table 1. Maximum mercury content of lamps

<i>Lamp</i>	<i>Maximum content</i>
T-8 fluorescent, eight-foot	10 mg mercury
T-8 fluorescent, four-foot	3.5 mg mercury
T-8 fluorescent, U-bent	6 mg mercury
T-5 fluorescent, linear	2.5 mg mercury
T-5 fluorescent, circular	9 mg mercury
Compact fluorescent, nonintegral ballast	3.5 mg mercury
Compact fluorescent, integral ballast	3.5 mg mercury, ENERGY STAR qualified
High-pressure sodium, up to 400 watts	10 mg mercury

High-pressure sodium, above 400 watts	32 mg mercury
---------------------------------------	---------------

mg = milligram

GUIDANCE

There are no substantive changes to the prerequisite requirements; refer to the LEED v4 reference guide, with the following addition:

Further Explanation

Connection to Ongoing Performance

- ▶ LEED O+M MR credit Purchasing: This credit is a strategy that can help achieve the MR credit: Purchasing in O+M v4.1 rating system.

MR Credit: Building Life-Cycle Impact Reduction

This credit applies to

- ▶ BD+C: New Construction (1-5 points)
- ▶ BD+C: Core & Shell (1-5 points)
- ▶ BD+C: Schools (1-5 points)
- ▶ BD+C: Retail (1-5 points)
- ▶ BD+C: Data Centers (1-5 points)
- ▶ BD+C: Warehouses & Distribution Centers (1-5 points)
- ▶ BD+C: Hospitality (1-5 points)
- ▶ BD+C: Healthcare (1-5 points)

INTENT

To encourage adaptive reuse and optimize the environmental performance of products and materials.

REQUIREMENTS

NC, CS, SCHOOLS, RETAIL NC, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY NC, HEALTHCARE

Demonstrate reduced environmental effects during initial project decision-making by reusing existing building resources or demonstrating a reduction in materials use through life-cycle assessment.

Achieve one or more of the following options below for a maximum of 5 points.

Option 1. Building and Material Reuse (1-5 points)

Maintain the existing building structure, envelope, and interior nonstructural elements. Reused or salvaged materials from off site that are incorporated into the building can also contribute to the credit calculations. However, reuse materials contributing toward this credit may not contribute toward MR Credit Material Disclosure and Optimization – Sourcing of Raw Materials.

Historic, abandoned, or blighted buildings: Portions of buildings deemed structurally unsound or hazardous can be excluded from the credit calculations.

Path 1 and 2 reward projects that reuse structural and/or nonstructural elements based on the project area. Path 1 and 2 can be combined for points.

Path 1: Maintain Existing Structural Elements: Walls, Floors, Roofs, and Envelope (1-5 points):

Maintain the existing building structure (including floor and roof decking) and envelope (the exterior skin and framing, excluding window assemblies and nonstructural roofing materials). Calculate reuse of the existing project area according to Table 1.

Table 1. Path 1 Points for reuse of existing building structural elements

<i>Percent of existing walls, floors and roof reuse by project area</i>	<i>Points BD+C</i>
15%	1
30%	2
45%	3
60%	4
75%	5

AND/OR

Path 2: Maintain Interior Nonstructural Elements (1 point)

Use existing interior nonstructural elements (e.g. interior walls, doors, floor coverings and ceiling systems) for at least 30% of the entire completed building, including additions.

AND/OR

Option 2. Whole-Building Life-Cycle Assessment (1-4 points)

For new construction (buildings or portions of buildings), conduct a cradle-to-grave life-cycle assessment of the project's structure and enclosure and select one or more of the following paths below to earn up to 4 points:

Path 1: Conduct a life cycle assessment of the project's structure and enclosure (1 point).

Path 2: Conduct a life cycle assessment of the project's structure and enclosure that demonstrates a minimum of 5% reduction, compared with a baseline building in at least three of the six impact categories listed below, one of which must be global warming potential (2 points).

Path 3: Conduct a life cycle assessment of the project's structure and enclosure that demonstrates a minimum of 10% reduction, compared with a baseline building, in at least three of the six impact categories listed below, one of which must be global warming potential (3 points).

Path 4: Meet requirements of Path 3 and incorporate reuse and/or salvage materials into the project's structure and enclosure for the proposed design. Demonstrate reductions compared with a baseline building of at least 20% reduction for global warming potential and demonstrate at least 10% reduction in two additional impact categories listed below (4 points).

For Paths 2, 3 and 4 listed above, no impact category assessed as part of the life-cycle assessment may increase by more than 5% compared with the baseline building. Include a narrative of how the life cycle assessment was conducted and if applicable for paths 2, 3 and 4 what changes were made to proposed buildings in order to achieve the related impact reductions.

The baseline and proposed buildings must be of comparable size, function, orientation, and operating energy performance as defined in EA Prerequisite Minimum Energy Performance. The service life of the baseline and proposed buildings must be the same and at least 60 years to fully account for maintenance and replacement. Baseline assumptions must be based on standard design and material selection for the project location and building type. Use the same life-cycle assessment

software tools and data sets to evaluate both the baseline building and the proposed building, and report all listed impact categories. Data sets must be compliant with ISO 14044.

Select at least three of the following impact categories for reduction:

- ▶ global warming potential (greenhouse gases), in kg CO₂e;
- ▶ depletion of the stratospheric ozone layer, in kg CFC-11e;
- ▶ acidification of land and water sources, in moles H⁺ or kg SO₂e;
- ▶ eutrophication, in kg nitrogen eq or kg phosphate eq;
- ▶ formation of tropospheric ozone, in kg NO_x, kg O₃ eq, or kg ethene; and
- ▶ depletion of nonrenewable energy resources, in MJ using CML / depletion of fossil fuels in TRACI.

GUIDANCE

Refer to the LEED v4 reference guide, with the following additions and modifications:

Behind the Intent

Beta Update

Updates in v4.1 include a greater focus on reducing embodied carbon of building structures and enclosures. Changes to this credit are intended to incentivize reuse of existing buildings and components, as well as make building reuse calculations simpler. Further, if buildings or building elements cannot be re-used significantly, changes to the lifecycle analysis option of the credit encourage projects to conduct whole building life cycle assessment as an integral design component for many more buildings.

Former LEED v4 credit Option 1 (Historic Building reuse) and Option 2 (Renovation of Abandoned and Blighted Building) have been combined into Option 1 for v4.1. The former Credit Option 3 (Building and Material Reuse) now is replaced with a consolidated option 1 and includes two pathways for calculating building reuse. These changes consolidate the calculation methodology for all types of reuse and restore the LEED v2009 credit pathways (MR c1.1: Building reuse – maintain existing walls, floors and roofs; and MR c1.2: Building reuse – maintain interior nonstructural elements) that provided simpler, more prescriptive calculations. Projects that have experienced difficulty in selecting and demonstrating a combination of structural and non-structural reuse in v4 now have the option of selecting either one or both types of re-use (structural and/or non-structural interior) depending on project scope.

Credit Option 2 (formerly Option 4), Whole Building Life Cycle Assessment, is now available to projects that pursue building reuse strategies that also wish to conduct an WBLCA for the project. The total points available cannot exceed 5 points, though projects can earn one Exemplary Performance point if they achieve reuse and/or WBLCA points from Option 1 and 2 and exceed 5 points in combination.

Credit Option 2 now has an entry pathway that rewards the effort to conduct a whole building life-cycle assessment without having to demonstrate specific impact reductions. Further, an incremental second point is now possible for showing 5% reductions compared to a baseline for three impact categories including global warming potential (GWP). The third point in this option remains unchanged to the v4 WBLCA credit requirements (10% reductions compared to the baseline in three impact categories including GWP). Finally, to greater reward embodied carbon reductions through reuse, project teams can now earn up to four points in Credit Option 4 by incorporating building element reuse and/or salvaged materials into a project's structure and enclosure and perform a life cycle assessment that shows at least a 20% reduction in global warming potential and at least 10% reduction in two other impact categories.

Step-by-Step Guidance

Refer to LEED v4 reference guide with the following additions and modifications.

Option 1: Building and Material Reuse

New general guidance for v4.1 applicable to Paths 1 and 2:

Identify non-structural and structural elements of the existing space (e.g. walls, floors, roofs, doors, floor coverings, ceiling systems, etc.) that can be retained separately and in combination to select best pathway/approach for re-use (Path 1 or Path 2). Include elements reused onsite and/or salvaged from offsite as part of the reuse calculations.

Projects that incorporate part of an existing building but do not meet the requirements for this credit may apply the reused portion toward the achievement of MR Credit Sourcing of Raw Materials. To apply the reused portion for the Sourcing of Raw Material credit, determine the cost of each material. This cost will be the actual cost paid or, if the material came from on-site, the replacement value. The replacement value can be determined by pricing a comparable material in the local market; exclude labor and shipping. If a project team receives a discount from a vendor, the replacement value should reflect the discounted price as opposed to the list value. When the actual cost paid for the reused or salvaged material is below the cost of an equivalent new item, use the higher value (actual cost) cost of the new equivalent item in the calculations. When the cost to reclaim an item found on-site is less than the cost of an equivalent new item, use the cost of the new item (or replacement cost).

Option 2: Whole Building Life Cycle Assessment

Collect information needed to perform life cycle assessment of structure and enclosure of the building.

Follow standard process associated with performing a typical whole building life cycle assessment. In general this can be broken down into: 1) Define goal and scope of assessment, 2) Collect information about materials and scenarios, 3) Perform calculations for impacts using reliable LCA assessment tools, 4) Understand and interpret results, and 5) Document process and produce detailed assessment reports.

Ensure that the scope of the analysis is a cradle-to-grave assessment which includes environmental impacts associated with the life-cycle stages for the building structure and enclosure. Follow the LEED v4 reference guide for minimum requirements for LCA related to products, functional equivalence, service life and system boundary under Step 1 with the following clarifications:

- ▶ The system boundary of the analysis must include a cradle to grave scope (modules A-D). However, some gaps in sub-modules may exist due to the materials or dataset chosen and design optimizations attempted for the project. Gaps in sub-modules are allowed so long as the system boundary in total encompasses a cradle-to-grave assessment. The required modules for a compliant whole building life-cycle analysis include:
 - Product stage: include modules A1-A3.
 - Construction process: include at least module A4.
 - Use Stage: Include at least one module from B1-B5 (structural materials without expected impacts from use, maintenance, repair or replacement over the building life are allowed to be excluded).
 - End of life stage: Include at least one module from C1-C4.
- ▶ If the project site boundary includes detached ancillary structures, such as parking structures or outbuildings, those structures must be included in the WBLCA calculations.
- ▶ For projects demonstrating impact reductions compared to a baseline: The LCA software or tool used for the baseline and proposed design must be the same, with the same modules and impact categories evaluated.
- ▶ Note that LCA software or tools must have ISO-14044-compliant data sets and conform to ISO 21931-2017 and/or EN 15978:2011 and their data must meet the requirements of ISO 21930-2017 and EN 15804. Typically, the software tool providers will document they meet these criteria in the LCA output report.

Additional Guidance for Whole Building LCA Tool Providers:

The system boundary of a whole building LCA must include modules A-D as defined in ISO 21930 and EN 15804. LCA tools must have ISO-14044-compliant data sets and conform to ISO 21931-2017 and/or EN 15978:2011. Further, the underlying data must meet the requirements of ISO 21930-2017 and EN 15804.

- ▶ If LCI data is not available for certain products, LCA tools can incorporate product EPD data into their ISO 14044 compliant LCI datasets as long as:
 - The EPD has not expired.
 - The EPD scenarios are representative of contemporary technologies and/or practice, and are relevant to the project location.
 - The EPD data reports all indicators and system boundary information required by WBLCA tools.
 - The EPD or LCA clearly indicates which product (manufacturer and product name) or geographical region it reflects in compared to industrywide results of a material available in the tool.

Further Explanation

Calculations

Option 1: Building and Material Reuse

The v4 calculation of formerly credit Option 3. Building and Material Reuse in LEED v4 reference guide has been changed. Steps 1 and 3 have been removed. Step 2 (Reuse off-site materials) remains in effect.

The LEED v4 Building and Material Reuse concept and calculation for “surface area and layers of reuse” is no longer utilized. Instead, project teams will calculate reuse as follows below for Path 1 and Path 2.

Path 1 Calculations: Maintain Existing Structural Elements: Walls, Floors, Roofs and Envelope

The reuse calculation is based on the surface areas of major existing structural and envelope elements per equation 3. Structural support elements such as columns and beams are considered part of the larger surfaces they support, so they are not quantified separately.

Prepare a spreadsheet listing all envelope and structural elements within the existing building prior to construction or renovation. Quantify each item, listing the square footage of both the existing area and the retained area. Determine the percentage of existing elements that are retained by dividing the square footage of the total retained materials area by the square footage of the total existing materials area. Include any salvaged or reused materials that were sourced off-site and integrated into the project as part of the reused area in the calculations.

Take measurements as if preparing a bid for construction of a building. For structural floors and roof decking, calculate the square footage of each component. For existing exterior walls and existing walls adjoining other buildings or additions, calculate the square footage of the exterior wall only and subtract the area of exterior windows and exterior doors from both the existing and the reused area tallies. For interior structural walls (e.g., shear walls), calculate the square footage of one side of the existing wall element. Table 1 provides an example of the calculations for Path 1.

Table 1. Sample Building Structure and Envelope Reuse Calculation for Path 1.

Structure/Envelope Element	Existing Area (sf)	Reused Area (sf)	Percentage Reused (%)
----------------------------	--------------------	------------------	-----------------------

Foundation/slab on Grade	11,520	11,520	100.0%
2nd floor Deck	11,520	10,000	86.8%
1st floor interior structural Walls	240	240	100.0%
2nd floor interior structural Walls	136	136	100.0%
Roof Deck	11,520	11,520	100.0%
South Exterior Wall (excl. windows)	8,235	8,235	100.0%
East Exterior Wall (excl. windows)	6,535	6,535	100.0%
West Exterior Wall (excl. windows)	6,535	5,820	89.1%
<i>Off-site salvage - brick façade for north exterior wall (excl. windows)</i>	8,235	5,500	66.8%
<i>Off-site salvage - structural beams for new stairwell</i>	650	650	100.0%
total	65,126	60,156	92.4%

Exclude the following items from this calculation: nonstructural roofing material, window assemblies, structural and envelope materials that are deemed structurally unsound, hazardous materials, and materials that pose a contamination risk to building occupants.

Equation 3. Percentage of existing building reuse – maintain existing structural elements:

$$\text{Existing building reuse} = \frac{\text{area reused on-site} + \text{area reused from off-site}}{\text{existing building area}} \times 100$$

Path 2 Calculations: Maintain Interior Nonstructural Elements

This pathway focuses on reuse of interior, nonstructural elements and compares the retained and reused elements with the total completed area of interior elements. It is not necessary to calculate the total area of existing interior nonstructural elements prior to demolition. Include any salvaged or reused materials that were sourced off-site and integrated into the project as part of the reused area in the calculations.

Take measurements as if preparing a bid for flooring, ceiling, or painting:

- ▶ Finished ceilings and flooring areas (tile, carpeting, etc.). Use square footage or square meters to determine area.
- ▶ Interior nonstructural walls. Determine the finished area between floor and ceiling and count both sides.
- ▶ Exterior structural and party walls. If the interior finishes (e.g., drywall and plaster) have been reused, count only one side.
- ▶ Interior doors. Count surface area once.
- ▶ Interior casework. Calculate the visible surface area of the assembly.

Include items that have been saved but may have been relocated, such as full-height demountable walls and doors that were rehung. Also include reused items purchased or sourced off-site from other buildings or projects, such as from salvage yards or donations.

Fixed items, such as nonstructural walls and doors, are included in this credit and count toward the percentage of reuse when they perform the same function (e.g., doors reused as doors). If materials are

used for another purpose (e.g., doors made into tables), they can count toward the achievement of MR Credit: Sourcing of Raw Materials, but they cannot count toward both credits.

Table 2 illustrates a spreadsheet for determining credit compliance. The total area of all new and existing building materials (following construction) is determined. The total area of only the existing and reused components is then entered. The sum of the existing materials is then divided by the sum of the total building materials to obtain the overall percentage of retained components. Since the overall percentage of reused nonstructural interior materials exceeds 30% of the total area of all nonstructural interior building materials, the project earns 1 point.

Table 2. Sample Interior Nonstructural Element Reuse Calculation for Path 2.

Interior Non-Structural Element	Total Area* (sf)	Existing/ Reused (sf)	Percentage Reused (%)
Gypsum Board Wall Partitions – full Height	5,400	2,500	46.3%
Gypsum Board Wall Partitions – Partial Height	650	650	100.0%
Carpeting	15,000	0	0.0%
Resilient flooring	350	350	100.0%
Ceramic tile	150	150	100.0%
Suspended ceiling systems	10,400	6,500	62.5%
Gypsum Board ceilings	350	350	100.0%
Interior Doors (Wood)	525	420	80.0%
Interior Windows / sidelights	56	56	100.0%
<i>Offsite salvage: interior Doors (Metal)</i>	86	86	100.0%
<i>Offsite salvage: interior casework / cabinetry</i>	235	235	100.0%
totals	33,202	11,297	34.0%
*note: the total area calculation includes both new and existing/reused materials.			

Determine the percentage of existing elements that are retained by dividing the total area of all retained interior nonstructural elements by the total area of interior nonstructural elements following Equation 4.

Equation 4: Percentage of existing building reuse – maintain interior nonstructural elements:

$$\text{Interior nonstructural reuse} = \frac{(\text{area of retained interior nonstructural elements} + \text{area of elements reused from offsite})}{\text{total area of interior nonstructural elements}} \times 100$$

Projects that incorporate part of an existing building for reuse but do not meet the requirements for Path 2 may apply the reused portion toward the achievement of MR Credit Construction and Demolition Waste Management. To do so, determine an approximate weight or volume for existing building elements and count them as waste diversion in the credit calculations.

Option 2 (Formerly Option 4): Whole Building Life Cycle Assessment

Refer to Option 4 in the LEED v4 reference guide with the following modifications and additions:

This option now has four thresholds outlined in four pathways. To achieve one point, the project must conduct a compliant lifecycle assessment of the project's structure and enclosure. Include structural materials from any detached or ancillary buildings in the WBLCA. To earn additional points, the proposed building must demonstrate a reduction in global warming potential and in two of five other impact categories when compared to a baseline building and include a narrative to explain the changes being made to proposed buildings in order to achieve the impact reductions (see credit requirements).

Developing an appropriate baseline building is necessary for a compliant WBLCA to show reductions in global warming potential and other impacts. Product and material environmental characteristics in the baseline building must reflect standard design practices and typical material selection choices for the project location and building type. For instance, assuming zero-percent recycled content in some metal products or concrete mixes does not reflect typical practice in North America. Project teams should look to common project types in the region and review industry resources in order to develop accurate baselines for claimed impact reductions. In documenting the credit, project teams will need to include a description of why the baseline structure and enclosure systems represent typical construction for the project, location, and building type.

Within Option 2, choose Path 1 (whole building life cycle analysis of the project) and/or Path 2, 3 or 4 (comparative whole-building life cycle analysis) as outlined in credit requirements. Note that for Path 1, project teams must complete a standard Whole Building Life Cycle Assessment (WBLCA) of the proposed design and report the impact categories in a WBLCA report, however there are no thresholds for reductions necessary to earn this point.

For Path 2, 3 and 4 as mentioned above, project teams must conduct a comparative WBLCA analysis and include a narrative summarizing differences between baseline and proposed building that contribute to the differences in LCA results. Points can be achieved as follows:

- ▶ 2 points – demonstrated impact reduction of at least 5% in Global Warming Potential and at least 2 other impact categories.
- ▶ 3 points – demonstrated impact reduction of at least 10% in Global Warming Potential and at least 2 other impact categories.
- ▶ 4 points – demonstrated impact reduction of 20% in Global Warming Potential and at least 10% in at least 2 other impact categories. This option must also incorporate reuse and/or salvaged materials as a part of the proposed structure or enclosure design. Strategies that offset significant amounts of embodied GWP include reuse of foundations, concrete structures, metal systems, and other high-impact structural materials. Reuse elements may be from on-site or off-site, and should be modeled following guidelines for small scale reuse (see Further Explanation, Small Scale Reuse in LEED v4), these guidelines should also be used for large scale reuse incorporation in Path 4 for LEED v4.1.

Life-cycle impact measures or indicators

Refer to LEED v4 reference guide with the following additions:

Reporting of impact category results: Report impacts in units of “per square foot” or “per square meter”, rounded to the nearest 10^{-4} for all six impact categories.

Required Documentation

Documentation requirements for former Option 1 (historic building reuse) and Option 2 (renovation of abandoned or blighted building) are now included under the restructured Option 1: Building and Material Reuse.

Documentation requirement for Option 1, Building and Material Reuse:

- ▶ Path 1: Structural and nonstructural reused elements table and calculations

- ▶ Path 2: Interior nonstructural reused elements table and calculations

Documentation requirement for Option 2, Whole Building LCA:

- ▶ WBLCA report for structure and enclosure of building

Documentation requirement for Option 2, Whole Building LCA Path 2, 3 and 4:

- ▶ WBLCA report that includes description of LCA assumptions, scope and analysis process for baseline building and proposed building, life cycle impact assessment summary showing outputs of proposed building with percent change from baseline building for all impact categories, and a narrative indicating which path was pursued and how reductions were achieved.

Exemplary Performance

- ▶ Option 1: Path 1: Reuse 90% of the building
- ▶ Option 2: Achieve Path 4 and show 40% reduction in GWP.
- ▶ Achieve more than 5 points from a combination of Option 1 and Option 2.

Connection to Ongoing Performance

- ▶ LEED O+M MR prerequisite Waste Performance: The concept above is a collection of strategies that can help achieve MR prerequisite Waste Performance in the O+M v4.1 rating system.

MR Credit: Environmental Product Declarations

This credit applies to

- ▶ BD+C: New Construction (1-2 points)
- ▶ BD+C: Core & Shell (1-2 points)
- ▶ BD+C: Schools (1-2 points)
- ▶ BD+C: Retail (1-2 points)
- ▶ BD+C: Data Centers (1-2 points)
- ▶ BD+C: Warehouses & Distribution Centers (1-2 points)
- ▶ BD+C: Hospitality (1-2 points)
- ▶ BD+C: Healthcare (1-2 points)

INTENT

To encourage the use of products and materials for which life-cycle information is available and that have environmentally, economically, and socially preferable life-cycle impacts. To reward project teams for selecting products from manufacturers who have verified improved environmental life-cycle impacts.

REQUIREMENTS

NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE

Achieve one or more of the options below, for a maximum of 2 points.

Option 1. Environmental Product Declaration (EPD) (1 point)

Use at least 20 different permanently installed products sourced from at least five different manufacturers that meet one of the disclosure criteria below. (10 different permanently installed products from three different manufacturers for CS and Warehouses & Distribution Centers).

- ▶ Life-cycle assessment and environmental product declarations.
 - Products with a publicly available, critically reviewed life-cycle assessment conforming to ISO 14044 that have at least a cradle to gate scope are valued as one whole product for the purposes of credit achievement calculation.
 - Product-specific Type III EPD -- Internally Reviewed. Products with an internally critically reviewed LCA in accordance with ISO 14071. Products with product-specific internal EPDs which conform to ISO 14025, and EN 15804 or ISO 21930 and have at least a cradle to gate scope are valued as one whole product for the purposes of credit achievement calculation.
 - Industry-wide Type III EPD -- Products with third-party certification (Type III), including external verification, in which the manufacturer is explicitly recognized as a participant by the program operator. Products with industry-wide EPDs, which conform to ISO 14025, and EN 15804 or ISO 21930 and have at least a cradle to gate scope are valued as one whole product for the purposes of credit achievement calculation.
- ▶ Environmental Product Declarations which conform to ISO 14025 and EN 15804 or ISO 21930 and have at least a cradle to gate scope.
 - Product-specific Type III EPD -- Products with third-party certification (Type III), including external verification and external critical review are valued as 1.5 products for the purposes of credit achievement calculation.

Option 2. Embodied Carbon/LCA Optimization (1 point)

Use products that have a compliant embodied carbon optimization report or action plan separate from the LCA or EPD. Use at least 5 permanently installed products sourced from at least three different manufacturers. Products are valued according to the table below.

Report type	Reference Document(s) for the Optimization Report	Report Verification	Valuation
Embodied Carbon/LCA Action Plan	Product-specific LCA or product-specific Type III EPD	Prepared by the manufacturer and signed by company executive	½ product
Reductions in Embodied Carbon: <10% reduction in GWP relative to baseline	Baseline: Product-specific LCA, Product-specific Type III EPD, or Industry-wide Type III EPD Optimized: Product-specific LCA or product-specific Type III EPD	Comparative analysis is verified by an independent party	1 product
Reductions in Embodied Carbon: 10%+ reduction in GWP relative to baseline			1.5 products
Reductions in Embodied Carbon: 20%+ reduction in GWP and 5%+ reduction in two additional impact categories, relative to baseline			2 products

Note: Reference documents for the optimization reports must be compliant with EPD Credit Option 1.

Impact categories:

- global warming potential (greenhouse gases), in CO₂e;
- depletion of the stratospheric ozone layer, in kg CFC-11e;
- acidification of land and water sources, in moles H⁺ or kg SO₂e;
- eutrophication, in kg nitrogen equivalent or kg phosphate equivalent;
- formation of tropospheric ozone, in kg NO_x, kg O₃ eq, or kg ethene; and
- depletion of nonrenewable energy resources, in MJ using CML / depletion of fossil fuels in TRACI.

For credit achievement calculation, products sourced (extracted, manufactured, purchased) within 100 miles (160 km) of the project site are valued at twice their base contributing number of products, up to a maximum of 2 products.

GUIDANCE

Refer to the LEED v4 reference guide, with the following additions and modifications:

Behind the Intent Beta Update

The prevalence of EPDs has grown significantly since LEED v4 was introduced in 2012. While uptake of Option 1 continues to increase in many regions, a significant number of product categories lack PCRs and many manufacturers do not have accompanying EPDs. Furthermore, Option 2 has been difficult to achieve due to the significant effort to investigate supply chains and the time it takes to reformulate products and develop comparable EPDs. Therefore, changes were made in v4.1 to simplify the selection

of products with life cycle assessment data and increase the ability to select optimized products in LEED.

For Option 1 of the EPD credit, partial weightings for life cycle assessment reporting formats have been removed to simplify project team selection and review of life cycle impact disclosures. All compliant product-specific LCAs and EPDs are now worth one whole product, with one exception: third party certified type III EPDs with external verification of the LCA are valued more (1.5 products) to reward highest quality EPDs. In addition, the number of products threshold for credit achievement has been reduced for BD&C - Core and Shell and Warehouse and Distribution Center projects because of their reduced project scopes. In all cases, the Exemplary Performance option for Option 1 has been updated to reflect these changes.

For Option 2, new pathways are intended to reward initial first steps, leadership in life cycle impact reductions, and optimized products that have lowered embodied carbon and other impacts. The cost threshold has been removed to allow simpler achievement criteria. Option 2 now only has a number of products-based threshold: procure 5 products from at least three different manufacturers. Manufacturers that do not have comparable lifecycle data will find an entry-level pathway in Option 2 that rewards “action plans” for those who demonstrate initiative towards reducing life cycle impacts.

Finally, the limitation previously set on contribution of structure and enclosure materials towards total percent by cost of eligible products has been removed.

Step-by-Step Guidance

Select which option(s) to pursue. Option 1 and 2 are now based on number of products only, and some products may contribute to both Options 1 and 2. Option 1 rewards the selection of products having product-specific declarations, industry wide EPDs, or product specific type III EPDs. Products must be sourced from multiple manufacturers as indicated in credit requirements.

Option 1: Environmental Product Declarations

Step 1: Specify and select compliant products

Follow LEED v4 reference guide for how to start specifying products with available life cycle assessments or environmental product declarations. Note that if a single EPD covers multiple formulations or product types but reports only combined impact results, that EPD can only count as 1 product.

Step 3: Count compliant products and materials and compile documentation

Follow LEED v4 guidance keeping in mind that products with product-specific LCAs or EPDs meeting more than one criterion are now all equally weighted at the same valuation factor of 1 product. However, Type III EPDs that include external verification of the EPD declaration and external verification of the underlying LCA data are valued as 1.5 products. A Type III EPD is considered externally verified if the person conducting the third-party verification of the underlying LCA data is independent and outside of the organization (as per ISO 14025 and EN 15804 or ISO 21930) in which the EPD is developed.

Also, note that the requirement for EPDs applies to the final unit of purchase—that is, entire product assemblies, not individual components. For example, a window shade that is only sold with attached metal hardware and mounting system cannot separately count the fabric, metal housing, or fasteners as individual EPDs. Or, a metal stud wall system that can only be installed with the appropriate clips and fastening system are considered one product because the system only functions as a whole.

Option 2: Embodied Carbon/LCA Optimization

Step 3: Calculate compliance

Option 1: With the data collected in a tracking tool or the calculator provided by USGBC, use Equation 1 (see *Further Explanation, Calculations*) to calculate the total number of products that comply with Option 1 requirements. This equation calculates compliance based on the number of products, not their cost. Product-specific LCAs and various types of compliant EPDs are valued at 1 product, with the exception of third-party externally verified EPDs that are valued at 1.5 products for credit achievement purposes.

Option 2: The cost-based pathway is no longer available. Choose products that sum to at least 5 products from 3 or more manufacturers that meet at least one of the requirements listed in Option 2. Track products using the credit calculator or an offline tool and calculate the number of products based on attributes and multipliers. Note: in v4.1 there is no limit/cap on structure and enclosure materials towards contributing to the value of compliant building products.

Further Explanation

Calculations

Refer to LEED v4 reference guide with the following modifications to the equations – Equation 1 for Option 1 and Equations 2 and 3 for Option 2 (note that calculations for LEED v4 Equation 3 for alternative structure and enclosure limit are no longer applicable to this v4.1 credit).

Equation 1: Total number of products with environmental product declarations (Option 1)

Total # of products = {# of products with product specific declarations/industry specific declarations/internally verified type III EPDs* X 1} + {# of Type III EPDs with external verification and external critical review X 1.5}

Equation 2: Percentage of cost for materials with Embodied Carbon/LCA Optimization Reports (Option 2)

The cost-based pathway is no longer used for this credit. See Equation 3 below.

Equation 3: Total number of products with Embodied Carbon/LCA Optimization Reports (Option 2)

Total # of products = {# of products with action plans X 0.5 X location valuation factor} + {# of products with any Third-Party verified impact reductions in GWP impact category X 1 X location valuation factor} + {# of products with Third-Party verified impact reductions with a minimum of 10% reduction in GWP impact category x 1.5 X location valuation factor} + {# of products with Third-Party verified impact reductions with a minimum of 20% reduction in GWP impact category and a minimum of 5% reduction in 2 other impact categories X 2 X location valuation factor}

Where,

- ▶ Location valuation factor = multiplier for the extraction, manufacture, and purchase location (see *MR Overview, Location Valuation Factor*).

Note: no single product may contribute more than 2X value.

Option 1: Additional Guidance for Type III EPDs for Manufacturers and Program Operators

Manufacturers and Program Operators: Third party certified (Type III) EPDs have been split into two categories for LEED v4.1: those with internal critical review, and those with external review and verification. LEED v4.1 introduces a new ISO standard to help guide best practices in critical review by referencing ISO 14071 which provides guidelines for conducting a critical review of any type of LCA study and the competencies required for the review. Any Type III EPD (whether internally or externally

reviewed) must follow the guidelines of ISO 14071 for reviewer and panelist qualifications and reporting consistency.

Externally critically reviewed and externally verified type III EPDs now are rewarded a multiplier of 1.5 “products” for credit calculation purposes. An EPD is considered externally verified if a person conducting the third-party verification is independent and outside of the organization (as per ISO 14025 and EN 15804 or ISO 21930) in which the EPD is developed.

Documentation of Product-Specific LCA Reports: Summary Table

Manufacturers and Program Operators: Product-specific declarations are defined for this credit as declarations that are based on a life-cycle assessment of a product but not constituting a full EPD (for EPD documentation requirements, see *Documentation of EPDs: Summary Table*). In addition to the documentation outlined in the LEED v4 Reference Guide, product-specific LCA declarations published after July 1, 2021 must include a cover sheet or summary table with the following information at a minimum:

- ▶ All requirements outlined in LEED v4 reference guide for this section
- ▶ The name/credentials of person(s) conducting the life cycle assessment
- ▶ The type of LCA software used to conduct the assessment;
- ▶ Date of assessment with period of validity or expiration date of life cycle assessment
- ▶ Contact information of the declaration holder or producer (typically the manufacturer)
- ▶ Unique document ID number of the report
- ▶ Product type
- ▶ Product name(s) and/or product line(s) covered by the report
- ▶ Product description
- ▶ Indication of impact categories measured
- ▶ Standards met
- ▶ Independent review entity’s name and statement
- ▶ Reference PCR
- ▶ The name/credentials of person(s) conducting the life cycle assessment
- ▶ URL link to the publicly available version of the document.

Documentation of EPDs: Summary Table

Manufacturers and Program Operators: In addition to the documentation outlined under *Required Documentation* in the LEED v4 Reference Guide, all EPDs published after July 1, 2021 must include a separate cover sheet or summary table. The summary table must include:

- ▶ All requirements outlined in LEED v4 reference guide for this section
- ▶ The name/credentials of person(s) conducting the life cycle assessment
- ▶ The type of LCA software used to conduct the assessment;
- ▶ Date of assessment with period of validity or expiration date of life cycle assessment,
- ▶ A reference to the valid PCR
- ▶ Names of global regions covered under the EPD
- ▶ URL link to the publicly available version of the document.
- ▶ Unique document ID number of the report
- ▶ Product type
- ▶ Product name(s) and/or product line(s) covered by the report
- ▶ Expiration date for the report

Option 2 Embodied Carbon/LCA Optimization Report Guidance - Action Plans Compliant Embodied Carbon/LCA Action Plans are standalone documents provided by manufacturers and valued at ½ product for credit calculation purposes. Action plans are created after a manufacturer has conducted a

life-cycle assessment of the product or product type, obtained results for the product in any of the life-cycle impact reporting formats per EPD Option 1, and has generated a publicly available narrative and action plan describing life-cycle assessment analysis with results.

Guidance for Manufacturers and Program Operators Creating Embodied Carbon/LCA Action Plan Reports:

A compliant Action Plan is developed by a manufacturer and based upon analysis from a product-specific LCA or product-specific type III EPD that uses EN 15804 or ISO 21930. The action plan must use the specified PCR functional unit for the LCA or EPD document. Action plans cannot be based on industry-wide LCA data or industry-wide EPDs and can be valid for no more than four years. Note: industry averages derived from a source other than an industry-wide EPD are not acceptable. Manufacturers using an industry-wide EPD must have been part of the industry-wide EPD in order to use it for optimization purposes.

The action plan must be publicly available and include strategies and analysis communicating how a manufacturer plans to mitigate or reduce life cycle impacts over time with special consideration of GWP reductions. The Action Plan does not have to be third party verified but at a minimum must be prepared by an individual with experience conducting product-specific LCAs. The Action Plan must be signed by an executive of the manufacturing company.

Action Plans must be a standalone document that includes the following information:

- ▶ Unique document ID number of the report
- ▶ Product type
- ▶ Product name(s) and/or product line(s) covered by the report
- ▶ Expiration date for the report
- ▶ Description of the LCA conducted including the dataset, software or platform used by manufacturer to complete the analysis.
- ▶ Provide a link to a summary of the underlying LCA data or EPD document.
- ▶ A table or otherwise summary of the largest life cycle impacts of their products throughout the product life cycle, which includes GWP impact result reporting at a minimum.
- ▶ Date of creation of the action plan as well as date of expiration (a maximum of four years for the plan).
- ▶ A written narrative describing immediate actions that will be pursued to reduce the overall life cycle impacts of the product within the four-year timeframe.
 - Include a description of the impact areas targeted for reduction in the action plan. The narrative shall describe how GWP is (or is not) targeted for reduction, including a numeric impact reduction target. Impact reductions must be shown as negative values and impact increases shown as positive values.
 - The narrative shall describe actions that will be pursued to reduce life cycle impacts of the product(s) within the four-year timeframe.
 - Include proposed changes in formulation or manufacturing processes that are planned as part of impact reduction strategy.
 - Include specific dates and a timeline for completion of all the steps described in the Action Plan. Only changes that are under the control of the manufacturer are valid for Action Plans.
 - Examples of optimization strategies under the control of the manufacturer include: expected changes from sourcing more local materials or suppliers to reduce shipping impacts, reduction in energy impacts from product manufacturing, an anticipation that the product will be designed to use less energy in the use phase or end of life phase, or installation of renewable energy for onsite generation at the manufacturing plant.
 - Examples of non-compliant changes are reductions that occurred without manufacturer action. For example, regulatory changes that result in lower impacts, energy grid emissions factor updates for the region, or LCA software

updates that result in lower impacts due to dataset changes rather than product optimizations.

- ▶ The reference LCA or EPD must meet all the requirements of the life cycle assessment reporting formats per Option 1 credit requirements and documentation requirements of product specific declarations or EPDs.

A sample of a compliant Embodied Carbon/LCA Action Plan is found in the *Resources* section of the online LEED credit library for this credit.

Option 2 Embodied Carbon/LCA Optimization Report Guidance

Project Team Members, Program Operators and Manufacturers: Compliant Embodied Carbon / LCA Optimization Reports are valued as follows for credit calculation purposes:

- ▶ Verified comparison documents that show any percent impact reduction in GWP via LCA or EPD compared with LCA or EPD (product-specific or industry-wide EPD if the manufacturer is recognized as a participant by the program operator), value at 1 product.
- ▶ Verified comparison documents that show 10% or more impact reduction in GWP via current LCA or EPD compared with previous LCA or EPD (product-specific or industry-wide EPD if the manufacturer is recognized as a participant by the program operator), value at 1.5 products.
 - Note: if using industry-wide EPDs or a product-specific LCA as the basecase for showing life cycle impact reductions in embodied carbon, the maximum value is 1.5 products.
- ▶ Verified comparison documents that show 20% or more impact reduction in GWP and at least 5% reduction in two additional impact categories via current product specific externally verified Type III EPD compared with a previous externally verified Type III EPD of same product based on same PCR, value at 2 products.
 - Since a 2X product multiplier is an exemplary level, only products that show improvement from a past product-specific Type III EPD to a newer product-specific Type III EPD can achieve the 2 products multiplier.

Guidance for Manufacturers and Program Operators Creating Embodied Carbon/LCA Optimization Reports:

To be eligible for showing reductions in life cycle impacts or embodied carbon for a product, a manufacturer must have previously conducted a product-specific LCA, product-specific Type III EPD, or participated in a published industry-wide Type III EPD in accordance with EPD Credit Option 1 Requirements. Then, the manufacturer has conducted a second life cycle assessment or published a second industry wide or product specific EPD in accordance to EPD Credit Option 1 Requirements for the same product type or product after making improvements towards impact reductions. Finally, the manufacturer has conducted a comparative analysis of the results between the two life cycle assessments per comparability guidelines in ISO 14025, section 6.7.2 or EN 21930, section 5.5. The comparison must be from two products following the same Product Category Rule of the same product function or product type. Industry averages derived from a source other than an industry-wide EPD are not acceptable. Manufacturers using an industry-wide EPD must have been part of the industry-wide EPD in order to use it for optimization purposes. Note that these requirements are for conducting a comparative analysis towards impact reductions and must not be used to make comparative assertions towards environmental claims for the product or product type.

The comparative analysis must show impact reduction in the global warming potential (GWP) impact category and must include a narrative describing how reductions in impacts were achieved. The published comparisons must be third-party verified. The comparative analysis and Embodied Carbon/LCA Optimization report can be developed by the same organization that conducted the newer

product-specific LCA or Type III EPD, but must be reviewed by an independent practitioner with experience in LCA.

The Embodied Carbon Optimization/LCA report must be a standalone document that includes the following information:

- ▶ Unique document ID number of the report
- ▶ Product type
- ▶ Product name(s) and/or product line(s) covered by the report
- ▶ Expiration date for the report
- ▶ Summary life-cycle impact category results from the baseline document (product-specific LCA, industry-wide Type III EPD, or product-specific Type III EPD). Results must be shown by module and impact category.
- ▶ Provide a link to the public version of the baseline document. Include the validation period and expiration date for the baseline document. The baseline document (LCA or EPD) cannot have expired more than 5 years before the newer LCA or EPD used for comparison was published.
- ▶ Summary life-cycle impact category results from the most current document (product-specific LCA or product-specific Type III EPD). Results must be shown by module and impact category. Provide a link to the public version of the current document. Include the validation period and expiration date for the newer document.
- ▶ LCA software details, LCA practitioner details and program operator details involved in development of both life cycle assessments.
- ▶ Expiration date of the comparative analysis (valid up to four years from when the second life-cycle assessment was conducted. The expiration date on the Optimization report should match the expiration date for the newest LCA or EPD document).
- ▶ The comparative analysis must show impact reduction in the global warming potential (GWP) impact category and include a narrative describing how reductions in impacts were achieved. Impact reductions must be shown as negative values and impact increases shown as positive values.
- ▶ Narrative explanation of the deliberate decisions taken to reduce life-cycle impacts of the product type or product.
- ▶ Only changes that are under the control of the manufacturer are valid for Optimization reports (i.e. “additional”):
 - Examples are expected changes from sourcing more local materials or suppliers to reduce shipping impacts, reduction in energy usage to manufacture the product within the manufacturing phase, an anticipation that the product will be designed to use less energy in the use phase, installation of renewable energy for onsite generation at the manufacturing plant, etc.
 - Examples of non-compliant changes are reductions that occurred without manufacturer action. For example, regulatory changes that result in lower impacts, energy grid emissions factor updates for the region, or LCA software updates that result in lower impacts due to the dataset rather than actual product optimizations.

Manufacturers and Program Operators: Third-Party Verification Program Requirements for Life Cycle Impact Reporting, Comparisons and Narrative:

- ▶ The third-party verification and certification program must conduct their operations in compliance with ISO 17065 or have been certified to ISO 17065.
- ▶ The verifier will confirm that the output document was conducted in compliance with the requirements as listed above.
- ▶ The third-party verifier of the life cycle comparisons and narratives must be different from the individual that created the LCA.
- ▶ The verifier must be qualified to conduct verification of life-cycle assessment reports, comparisons and narrative.
- ▶ The verifier must receive regular training on the verification process and updates to the program regularly to ensure consistency of verification.
- ▶ The program must have a database of publicly available third-party verified products to meet these requirements as listed above.

- ▶ The program must have a written third-party verification process and procedure that is updated regularly with a process to implement updates and changes to verifiers of the program that is publicly available.

Notes on Calculations:

- ▶ No single product may contribute more than 2X value.
- ▶ All product reports must be valid at the time the product was purchased for the project.

Required Documentation

Follow LEED v4 reference guide documentation requirements with the following modifications for Option 2:

- ▶ Option 1: use the LEED v4.1 MR building product calculator. For products that have been verified for LEED by GBCI, include “Verified for LEED Documentation” ID numbers, or copies of EPD/LCA reports and compliant summary documents in LEED Online for products contributing toward credit.
- ▶ Option 2: use the LEED v4.1 MR building product calculator. For products that have been verified for LEED by GBCI, include “Verified for LEED Documentation” ID numbers for products selected in the calculator, or copies of action plans and optimization reports in LEED Online for products contributing toward credit.

Exemplary Performance

Option 1: Purchase at least 40 qualifying permanently installed building products from ten different manufacturers that meet the credit criteria (Purchase at least 20 products from five different manufacturers for Core and Shell and Warehouse & Distribution Centers).

Option 2: Purchase at least 10 compliant products from five different manufacturers of permanently installed building products that meet the credit criteria.

Connection to Ongoing Performance

- ▶ LEED O+M MR prerequisite Waste Performance, MR prerequisite Purchasing Policy, and MR credit Purchasing: A similar credit having some of the above requirements for existing buildings is required for the O+M v4.1 rating system and is a strategy that can help achieve MR prerequisite Waste Performance, MR prerequisite Purchasing Policy, and MR credit Purchasing.

MR Credit: Sourcing of Raw Materials

This credit applies to

- ▶ BD+C: New Construction (1-2 points)
- ▶ BD+C: Core & Shell (1-2 points)
- ▶ BD+C: Schools (1-2 points)
- ▶ BD+C: Retail (1-2 points)
- ▶ BD+C: Data Centers (1-2 points)
- ▶ BD+C: Warehouses & Distribution Centers (1-2 points)
- ▶ BD+C: Hospitality (1-2 points)
- ▶ BD+C: Healthcare (1-2 points)

INTENT

To encourage the use of products and materials for which life cycle information is available and that have environmentally, economically, and socially preferable life cycle impacts. To reward project teams for selecting products verified to have been extracted or sourced in a responsible manner.

REQUIREMENTS

NC, CS, SCHOOLS, RETAIL NC, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY NC, HEALTHCARE

Responsible Sourcing of Raw Materials (1-2 points)

Use products sourced from at least three different manufacturers that meet at least one of the responsible sourcing and extraction criteria below for at least 15%, by cost, of the total value of permanently installed building products in the project (1 point).

Use products sourced from at least five different manufacturers that meet at least one of the responsible sourcing and extraction criteria below for at least 30%, by cost, of the total value of permanently installed building products in the project (2 points).

- ▶ *Extended producer responsibility.* Products purchased from a manufacturer (producer) that participates in an extended producer responsibility program or is directly responsible for extended producer responsibility. Products meeting extended producer responsibility criteria are valued at 50% of their cost for the purposes of credit achievement calculation.
- ▶ *Bio-based materials.* Bio-based products and materials other than wood must be tested using ASTM Test Method D6866 or equivalent method ISO 16620-2, or be certified to the USDA BioPreferred Voluntary Labeling Initiative that includes verification via ASTM 6866 testing. Exclude hide products, such as leather and other animal skin material.
 - Bio-based products that meet the criteria above: value at 50% of cost multiplied by the biobased content of the product for the purposes of credit achievement calculation.
- ▶ *Wood products.* Wood products must be certified by the Forest Stewardship Council or USGBC-approved equivalent. Products meeting wood products criteria are valued at 100% of their cost for the purposes of credit achievement calculation.
- ▶ *Materials reuse.* Reuse includes salvaged, refurbished, or reused products. Products meeting materials reuse criteria are valued at 200% of their cost for the purposes of credit achievement calculation.
- ▶ *Recycled content.* Products meeting recycled content criteria are valued at 100% of their cost for the purposes of credit achievement calculation.

- Recycled content is the sum of postconsumer recycled content plus one-half the preconsumer recycled content, based on weight.
- The recycled fraction of the assembly is then multiplied by the cost of assembly to determine the recycled content value.

For credit achievement calculation, products sourced (extracted, manufactured and purchased) within 100 miles (160 km) of the project site are valued at twice their base contributing cost, up to a maximum of 200% of cost.

GUIDANCE

Refer to the LEED v4 reference guide, with the following additions and modifications:

Behind the Intent

Beta Update

Option 1 of this credit in LEED v4 was intended to focus on improving the knowledge gaps and encourage best practices in raw materials sourcing and extraction. However, lack of specific enough Corporate Social Responsibility (CSR) reports have made achievement of this credit challenging. The intent of this credit option is still a priority and USGBC plans to move Option 1: Raw Material Source and Extraction Reporting to the Pilot Credit Library where it can be refined and updated.

By removing Option 1, Option 2 is now worth two points in version 4.1, and the title has changed to “Responsible Sourcing of Raw Materials.” Minor updates have been made to the credit criterion to better reward embodied carbon, renewable bio-based materials, and foster circular economies through recycling and reuse. Finally, the limitation on structure and enclosure materials has been removed. These changes are intended to increase uptake of these important material attributes and continue our efforts to refine best practices in materials extraction and sourcing.

Step-by-Step Guidance

For v4.1, projects no longer have to choose between the raw material source reporting option (previously Option 1) and optimized sourcing option (previously Option 2), hence projects should follow LEED v4 reference guide exclusively for leadership extraction practices (henceforth referred to as responsible sourcing of raw materials) and choose between the two new thresholds of 15% by cost (for 1 point) or 30% by cost (for 2 points) for the sourced products.

Further Explanation

Calculations

Equation 1 and Equation 3 (for alternative structure and enclosure limit) in the Reference Guide are no longer applicable to this credit. Instead, use Equation 2: Percentage of responsibly sourced products for applicable product cost, criterion valuation factor and location valuation factor with the following modifications in criterion valuation factor of different sourcing attributes:

Product cost = cost of product contributing toward credit.

Criterial valuation factor = multiplier assigned to each sourcing criterion:

- ▶ Bio-based products meeting basic criteria, value 0.5, by cost; bio-based products meeting Sustainable Agriculture Standard, value 1.0 by cost.
- ▶ Wood products certified to FSC standards, value 1.0 by cost (no change).
- ▶ Reused materials, value 2.0 by cost. Calculate cost according to the MR Credit Building Lifecycle Impact reduction, Option 3.
- ▶ Recycled content:
 - Postconsumer recycled materials, value 1.0 by cost (no change)

- Pre-consumer recycled materials, value 0.5 by cost (no change)
- Recycled content is the sum of postconsumer recycled content plus one-half the pre-consumer recycled content, based on weight. The recycled fraction of the assembly is then multiplied by the cost of assembly to determine the recycled content value.
- ▶ Location valuation factor: multiplier for extraction, manufacture and purchase location is 200% (no change)
- ▶ Extended Producer Responsibility is valued at 50%; valuation factor is 0.5 (no change)

Extended Producer Responsibility (EPR)

Follow LEED v4 reference guide with the following additional specifications for the two basic types of EPR programs:

- ▶ EPR claims must be made in accordance to ISO 14021.
- ▶ The manufacturer and/or programs have provided documentation showing participation in any of the following EPR platforms to contribute to LEED:
 - Manufacturer Based Programs:
 - A narrative from the manufacturer describing the nature of the program, where the materials can be collected, and where the materials go upon collection.
 - A state recognized program.
 - Third-party program:
 - A narrative from a manufacturer or third-party entity that includes language on how the third-party is directly responsible for the take back of materials. Within this narrative includes:
 - Collection facility locations.
 - Description of how materials are processed.
 - Fate of materials after they are processed.

Documentation for wood and bio-based products:

- ▶ Products with bio-based claims must be legally harvested by exporting and receiving country.
- ▶ Manufacturers/Programs must provide documentation on the specific product that includes the type of bio-based raw material used within the product. In addition, products must have either of the following:
 - Confirmation the ASTM D6866 test method was conducted which validates the percent by weight of bio-based material within the product.
 - Product has been certified to the USDA BioPreferred Program and includes the Voluntary Labeling verification of bio-based content via ASTM 6866 or ISO 16620-2 testing.

About the BioPreferred Voluntary Labeling initiative:

Some products listed on the United States Department of Agriculture's BioPreferred Program Catalog include a voluntary labeling logo that indicates the biobased content of the product has been verified per ASTM D6886 testing. www.biopREFERRED.gov/BioPreferred

Calculating bio-based material contributions

- ▶ Calculate bio-based content using this formula:
 - $50\% \times \text{percent by weight of bio-based material in total product} \times \text{Cost of Material}$

Recycled Content: Follow LEED v4 reference guide with the following additional specifications:

Calculating recycled content contributions:

- ▶ Calculate product recycled content using this formula:
 - $[\% \text{ Pre Consumer} \times 50\%] + [\% \text{ Post Consumer} \times 100\%] \times \text{Cost of Material}$

Notes on Calculations:

- ▶ No single product may contribute more than 200% of cost.
- ▶ All product reports must be valid at the time the product was purchased for the project.

Required Documentation

Follow LEED v4 reference guide documentation requirements with the following modifications:

- ▶ Documentation requirements for Option 1 are no longer applicable.
- ▶ Instead, submit the LEED v4.1 MR building product calculator and documentation of product claims for credit requirements for either 15% by cost or 30% by cost threshold.
- ▶ For products that have been verified for LEED by GBCI, include ID numbers in the building product calculator. Additional documentation is not necessary to be included to LEED Online.

Exemplary Performance

Purchase at least 45%, by cost, of the total value of permanently installed building products that meet the credit criteria.

Connection to Ongoing Performance

- ▶ LEED O+M MR prerequisite Waste Performance, MR prerequisite Purchasing Policy, and MR credit Purchasing: A similar credit having some of the above requirements for existing buildings is required for the O&M v4.1 rating system and is a strategy that can help achieve MR prerequisite Waste Performance, MR prerequisite Purchasing Policy, and MR credit Purchasing.

MR Credit: Material Ingredients

This credit applies to

- ▶ BD+C: New Construction (1-2 points)
- ▶ BD+C: Core & Shell (1-2 points)
- ▶ BD+C: Schools (1-2 points)
- ▶ BD+C: Retail (1-2 points)
- ▶ BD+C: Data Centers (1-2 points)
- ▶ BD+C: Warehouses & Distribution Centers (1-2 points)
- ▶ BD+C: Hospitality (1-2 points)
- ▶ BD+C: Healthcare (1-2 points)

INTENT

To encourage the use of products and materials for which life-cycle information is available and that have environmentally, economically, and socially preferable life-cycle impacts. To reward project teams for selecting products for which the chemical ingredients in the product are inventoried using an accepted methodology and for selecting products verified to minimize the use and generation of harmful substances. To reward raw material manufacturers who produce products verified to have improved life-cycle impacts.

REQUIREMENTS

NC, CS, SCHOOLS, RETAIL NC, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY NC, HEALTHCARE

Option 1. Material Ingredient Reporting (1 point)

Use at least 20 different permanently installed products from at least five different manufacturers that use any of the following programs to demonstrate the chemical inventory of the product to at least 0.1% (1,000 ppm). (10 different permanently installed products from at least three different manufacturers for CS and Warehouses & Distribution Centers)

- ▶ *ANSI/BIFMA e3 Furniture Sustainability Standard.* The documentation from the assessor or scorecard from BIFMA must demonstrate the product earned 4, 5, 7, or 8 points under 7.5.1.1 Chemical Assessment in e3-2019 (Pathway 1), 3 points under 7.5.2.2 Advanced Level in e3-2019 (Pathway 2), or at least 3 points under 7.5.1.3 Advanced Level in e3-2014.
 - For e3-2019: If product achieved 3 points under 7.5.1.1 in e3-2019 using the GHS classification sub-path, then the product meets this requirement. Manufacturer to provide additional backup documentation to show which sub-path was used in Pathway 1 (7.5.1) in this instance.
- ▶ *Cradle to Cradle.* Product has Material Health Certificate or is Cradle to Cradle Certified™ under standard version 3 or later with a Material Health achievement level at the Bronze level or higher.
- ▶ *Declare.* The Declare product label must meet the following requirements:
 - Declare labels designated as Red List Free, LBC Red List Free, or Declared.
 - Declare labels designated as LBC Red List Approved or LBC Compliant that demonstrate content inventory to 0.1% (1,000 ppm).
- ▶ *Facts - NSF/ANSI 336:* Sustainability Assessment for Commercial Furnishings Fabric at any certification level.
- ▶ *Global Green TAG.* Product Health Declaration (PHD) labels issued after January 1, 2020.
- ▶ *Health Product Declaration.* The end use product has a published and complete Health Product Declaration with full disclosure of known hazards in compliance with the Health Product Declaration Open Standard.
- ▶ *Living Product Challenge.* The included Declare product label must demonstrate content inventory to 0.1% (1,000 ppm).

- ▶ *Manufacturer Inventory.* The manufacturer has published complete content inventory for the product following these guidelines:
 - A publicly available inventory of all ingredients identified by name and Chemical Abstract Service Registration Number (CASRN) and/or European Community Number (EC Number).
 - Materials defined as trade secret or intellectual property may withhold the name and/or CASRN/EC Number but must disclose ingredient/chemical role, amount and hazard score/class using either:
 - Greenscreen List Translator (LT) score and/or Full GreenScreen Benchmark (BM)
 - The Globally Harmonized System of Classification and Labeling of Chemicals rev.6 (2015) (GHS)
 - The hazard screen must be applied to each trade secret ingredient and the inventory lists the hazard category for each of the health hazards included in Part 3 of GHS (e.g. “GHS Category 2 Carcinogen”).
- ▶ *Product Lens Certification*

Any compliant reports above with third-party verification that includes the verification of content inventory are worth 1.5 products for credit achievement calculations.

AND/OR

Option 2: Material Ingredient Optimization (1 point)

Use products that have a compliant material ingredient report or action plan. Use at least 5 permanently installed products sourced from at least three different manufacturers. Products are valued according to the table below.

Report Type & Criteria	Product Documentation	Report Verification	Valuation
Material Ingredient Screening and Optimization Action Plan	Action Plan based on publicly available material inventory to at least 1,000ppm.	Prepared by the manufacturer and signed by company executive	½ product
Advanced Inventory & Assessment: Inventory to at least 0.01% by weight (100 ppm) and no GreenScreen LT-1 hazards or GHS Category 1 hazards are present. Or Inventory to at least 0.01% by weight (100ppm) and at least 75% by weight of product is assessed using GreenScreen. The remaining 25% by weight of product has been inventoried and the GreenScreen assessment is publicly available.	<i>Cradle to Cradle Certified</i> or <i>Material Health Certificate</i> at Bronze level or higher. <i>Declare</i> labels designated as <i>Red List Free</i> or <i>LBC Red List Free</i>. <i>Green Seal</i>. Products certified under the Standard for Paints, Coatings, Stains and Sealers (GS-11, Edition 4.0) that do not include GHS Reproductive toxins (categories 1 and 2). <i>Health Product Declaration</i> that meet optimization and verification criteria. <i>Living Product Challenge</i> certified products that include a Red List Free or LBC Red List Free <i>Declare</i> label. <i>Manufacturer Inventory</i> that meet optimization and verification criteria.	Third-party verified	1 product
Material Ingredient Optimization: Inventory to at least 0.01% by weight (100 ppm) and at least 95% by weight of product is assessed using GreenScreen. No BM-1 hazards are present. The remaining 5% not assessed has been inventoried and screened using GreenScreen List Translator and no GreenScreen LT-1 hazards are present.	<i>Cradle to Cradle Certified</i> or <i>Material Health Certificate</i> at Silver level or higher. <i>Health Product Declaration</i> that meet optimization and verification criteria. <i>Living Product Challenge</i> certified products that achieve Imperative 09: Transparent Material Health. <i>Manufacturer Inventory</i> that meet optimization and verification criteria.		1.5 products
International Alternative Compliance Path: <i>Available to projects located outside of the US</i>	<i>REACH Optimization</i>: Material Inventory to 100ppm with no substances found on the Authorization List – Annex XIV, the Restriction list – Annex XVII and the SVHC candidate list. <i>Global Green TAG</i> PHD report.	REACH report prepared by the manufacturer PHD Report verified by <i>Global Green TAG</i>	1 product

For credit achievement calculation, products sourced (extracted, manufactured, purchased) within 100 miles (160 km) of the project site are valued at twice their base contributing number of products), up to a maximum of 2 products.

GUIDANCE

Refer to the LEED v4 reference guide, with the following additions and modifications:

Behind the Intent

Beta Update

Changes were made in v4.1 to reduce barriers for achievement while maintaining the overall approach to the Materials Ingredients credit:

- ▶ Option 1 maintains the threshold of 20 products for most BD+C project types but now will allow project teams to meet the credit via 10 products instead of 20 for less material intensive BD+C-C&S and BD+C-warehouse projects. Products with Option 1 compliant reporting methodologies or labels with third-party verification of content inventory will be worth 1.5 products.
- ▶ The cost-based threshold for Option 2 has been removed to simplify credit achievement.
- ▶ A restructuring of Option 2 rewards manufacturers at multiple steps along the path towards product material ingredient optimization. The new format provides a stepwise approach towards leadership for materials ingredient screening, assessment and optimization.
- ▶ Option 3: Product Manufacturer Supply Chain Optimization has been removed from the rating system and pilot credit requirements as well as guidance moved to the Pilot Credit Library where it will be refined for possible incorporation back into the rating system in the future.
- ▶ The limitation previously set on structure and enclosure materials for cost calculation has been removed.

Step-by-Step Guidance

Follow LEED v4 reference guide with the following modifications:

Option 2 Material ingredient optimization requires the selection of at least 5 products to meet at least one of the paths listed in the credit requirement. Option 3 requirements and guidance are no longer applicable.

Option 1. Material Ingredient Reporting

Step 2. Specify and select compliant products

Specify at least 20 products, from at least five different manufacturers for BD&C projects (with the exception of BD&C- Core and Shell and Warehouse projects, for these project types only 10 products from three different manufacturers need to be specified).

Option 2. Material Ingredient Optimization

Step 3: Calculate number of compliant products

Equation 2 (formerly for product supply chain optimization) is no longer applicable. Therefore, determine the total number of compliant products using the new Equation 2 below. LEED v4 Equation 3 for alternative structure and enclosure percentage limit is no longer applicable.

Further Explanation

Refer to LEED v4 reference guide with the following modifications (Note that LEED v4 Equations 2 (supply chain optimization) and 3 (Alternative structure and enclosure limit) are no longer applicable for use in the newer credit, and Equation 2 for v4.1 Material Ingredients credit refers to newer product number metric for Option 2.

Calculations

Use the new Equation 2 below for Option 2 Material Ingredient Optimization (also see *Further Explanation, Material Ingredient Optimization*)

Equation 2: Total number of products (Option 2)

Total # of products = {# of products with Action Plans X 0.5 X location valuation factor} + {# of products with materials meeting the requirements in section Advanced Inventory and Assessment X 1 X location valuation factor} + {# of products with materials meeting the requirements in section Material Optimization x 1.5 X location valuation factor}

Where,

- ▶ Location valuation factor = multiplier for the extraction, manufacture, and purchase location (see *MR Overview, Location Valuation Factor*).

Note: no single product may contribute more than 2X the product valuation.

Option 1: Material Ingredient Reporting – Additional Guidance for Manufacturers

Manufacturers: Refer to LEED v4 reference guide for this section with the following modifications and additions:

Cradle to Cradle (C2C) Certified and C2C Material Health Certificate

Refer to LEED v4 reference guide for this section with the following modifications:

The credit requirements for v4.1 are aligned with the latest version of the C2C standard v3 and v3.1. C2C v2.1.1 is no longer applicable.

The C2C Material Health Certificate uses material health assessment methodology of the C2C certified product standard to encourage awareness of chemicals in products and supply chains and contribute more specifically towards safer chemicals/greener chemistry within the C2C program. C2C Material Health Certificate can be obtained individually or as part of the C2C program and the requirements for both are identical. Both C2C Certified and C2C Material Health assessment are third-party verified claims that count as 1.5 products for the purpose of LEED v4.1 credits.

Declare and Living Product Challenge labels

Declare and Living Product Challenge labels are issued by the International Living Future Institute (ILFI) and generated via ILFI's Declare portal website or other tools. While third-party verification of Declare labels is not required in LEED, Declare labels that are content verified by a valid third party process are valued at 1.5 products for credit achievement calculations.

There are three kinds of Declare labels:

- ▶ “Red List Free,” also known as “LBC Red List Free”
- ▶ “LBC Compliant,” also known as “LBC Red List Approved”
- ▶ “Declared”

Products that receive the status of “Red List Free” or “LBC Red List Free” are 100% disclosed down to 100ppm level and do not contain any Red List chemicals. Products that receive the status of “Declared” are also 100% disclosed down to 100ppm level, but may contain one or more Red List ingredients that are not covered by an existing LBC temporary exception. All building products carrying “LBC Red List Free,” “Red List Free,” and “Declared” status meet LEED v4.1 requirements for disclosure. However, some products that receive the status of “LBC Compliant” or “LBC Red List Approved” may rely on one or more exceptions allowed in the program that may not satisfy the requirements for LEED credit achievement purposes. For example, some products that utilize Temporary Exception I10-E4 may

withhold some proprietary ingredient information of up to 1% of product ingredients (therefore not meeting the LEED disclosure threshold), while some other LBC compliant products may use that exemption to exclude or withhold less than 0.1% of product ingredients information (and therefore meet the requirements of the LEED credit). Product specifiers must closely review the labels to ensure they meet the LEED requirement. The Declare product directory has a filter for LEED v4 compliance. <https://living-future.org/declare/>

The *Living Product Challenge*, version 1.1 or version 2.0, includes a verified “Red List Free” or “LBC Compliant” Declare label as part of certification. Therefore, “*LBC Compliant*” labels may be able to withhold more than 1,000ppm due to temporary exceptions allowed in LBC Complaint labels, making their disclosure potential identical to those found in Option 1. However, any product meeting the Living Product Challenge must have 100% of the content inventory reviewed by an assessor to 100ppm and screened against the Red List and GreenScreen List Translator. For a product to achieve the Transparent Material Health Imperative within the Living Product Challenge, an assessor must assess a minimum of 95% of the product content. The remaining 5% must be screened against the Green Screen List Translator and the ILFI Red List. Therefore, any Living Product Challenge that achieves the Transparent Material Health imperative meets the Optimization thresholds for Option 2 of the credit.

Global GreenTag

Global GreenTag provides a variety of product certifications in North America and over 70 countries worldwide. Global Green Tag’s Product Health Declaration (PhD) label provides disclosure of contents, provides an evaluation of material ingredients, includes a list of banned ingredients, and provides evaluation of potential exposure and risks over the lifecycle stages of a product. PhD labels published after January 1, 2020 have a disclosure reporting format that is compliant with credit Option 1. Compliant PhD labels are worth 100% by cost, or 1 product, under Option 1.

Products that utilize ingredients found on the REACH Authorization, Restriction, and Candidate lists are not eligible for PhD certification, therefore PhDs are deemed to comply with the credit Option 2, International Alternative Compliance Path – REACH Optimization. PhDs are applicable for all projects outside the US where the REACH Optimization pathway is eligible on LEED projects. PhD reports are worth 100% by cost, or 1 product, under Option 2.

Green Seal

As a multi-attribute standard, Green Seal’s paints and coatings standard includes criteria for material health optimization in addition to criteria for efficacy, sustainable packaging, and labeling. It also includes stringent limits for VOC content and emissions that meet LEED v4.1 IEQ credit: Low-Emitting Materials requirements.

GreenSeal standard for Paints, Coating, Stains, and Sealers (GS-11, Edition 4.0) has been approved for Material Ingredients - Option 2. Certified products that meet LEEDv4.1 optimization criteria [i.e. they do not include GHS Reproductive toxins (categories 1 and 2)] are eligible to be counted as 1 product. The GS-11 standard establishes environmental, health, and performance requirements for certain architectural coatings that are intended to be applied on-site, and for stains, finishes, and sealers.

Find compliant products at: <https://certified.greenseal.org/directory>

Health Product Declaration Open Standard

Follow LEED v4 reference guide for this section with the following modifications and additions:

HPDs can now be generated via the HPD online builder available on the HPDC website. While third-party verification of HPDs is not required in LEED, HPDs that are content verified by a valid third party process are valued at 1.5 products. The third party verification status of HPDs can be confirmed on the summary page of a published HPD.

Qualifying HPDs developed under the Open Standard version 2.0, 2.1, 2.2 or subsequent versions of the standard are eligible for documenting credit achievement if the HPD is still valid at the time that the product is purchased and used on a project. Version 1.0 HPDs are currently expired in the marketplace and will not contribute to this LEED v4.1 credit unless compliant products were purchased during the time the HPD was valid.

Manufacturer Inventory

Follow LEED v4 reference guide for this section with following modifications:

If the specific ingredient cannot be disclosed for proprietary reasons, the manufacturer may withhold the name and CASRN or EC number but still provide ingredient role/function in product, amount as a percent of total product content (or ppm), and hazard score/class using Green Screen (GS) List Translator, GS Benchmark, or Globally Harmonized System (GHS) for Classification and Labeling of Chemicals v2015. Report hazard levels and hazard endpoints that result in scoring the ingredient as Benchmark 1 using full Green Screen. It is not necessary to report hazards associated with higher Benchmark levels or LT-UNK (Unknown) using GS List Translator.

- ▶ For reporting of proprietary ingredients in a manufacturer inventory via GHS pathway (Global Harmonized System of Classification and Labeling of Chemicals Category rev. 6 or higher)
 - The hazard screen must be applied to each trade secret ingredient and the inventory lists the hazard category for each of the health hazard included in Part 3 of GHS (e.g. “GHS Category 2 Carcinogen”).
 - Identify in the inventory all hazard classes for which a classification cannot be made because there is insufficient data for a particular endpoint (data gaps):
 - For a product manufacturer, this is like GS-LT UNK per the GreenScreen LT pathway.
 - A GHS compliant manufacturer inventory will specify either a category hazard for the substance or state there is insufficient data for the particular endpoint in a statement “insufficient data” next to the chemical.

In addition to the requirements above, compliant manufacturer inventories must follow the most recent version of the HPDC Open Standard for reporting, ingredient categorization, disclosure thresholds, displaying content ranges, hazards, and other special conditions.

Option 2: Material Ingredient Optimization Additional Guidance

Manufacturers: Refer to LEED v4 reference guide for this section with the following modifications and additions:

Under the newly structured Option 2, several different pathways can be used as starting points or frameworks for documenting substitution of problematic substances such as the screening and optimization action plan, manufacturer inventory, HPD (material transparency initiative from HPDC), Declare (product transparency label from ILFI), Green Screen (a program of Clean Production Action), Cradle to Cradle Certified and the European Union’s REACH program (for international projects only).

Option 2 goes beyond Option 1’s reporting requirement and encourages the use of products that have extensive screening and inventory of their ingredients as well as thorough assessment for potential health impacts and optimization of the ingredient chemistry. In order to achieve these goals, Option 2 has been restructured into three main compliance pathways: Material Ingredient Screening and

Optimization Action Plan (valued at 50% by cost or 0.5 product), Advanced Inventory and Assessment (valued at 100% by cost or 1 product) and finally, Material Ingredient Optimization (valued at 150% by cost or 1.5 products).

Additional GreenScreen related guidance for Option 2 with GreenScreen Benchmark 1 for manufacturers and suppliers

Refer to LEED v4 reference guide for this section with the following modifications:

Under Option 2 in LEED v4 previously, the two levels of compliance—GS List Translator and GS full assessment—were weighted at 100% of cost and 150% of cost, respectively. The newer requirements for Option 2 in LEED v4.1 (Advanced Inventory and Assessment and Material Ingredient Optimization) place a more deliberate focus on the concept of ingredient assessment beyond screening, now explicitly reward GS Benchmark assessments for 75% by weight of product (valued at 1 product) or for 95% by weight of product (valued at 1.5 product) and require the remaining percentage by weight product in both cases to be screened using GS List Translator.

In general, GreenScreen Benchmarks are defined for single chemicals and well-defined mixtures of ingredients, such as metal alloys. However, complex materials and those with unknown or variable composition are less suitable for being assigned a single GreenScreen Benchmark. Examples of such materials include wood, stone, sand, aggregate, and mixed recycled content materials. Building products using significant amounts of these materials may face challenges in screening against GreenScreen Benchmarks for these ingredients while seeking to satisfy the LEED credit requirements. The product certifications and declaration programs recognized in LEED for this credit have requirements for accommodating complex or variable composition materials. Where no such guidance exists, such as for the Manufacturer Inventory pathway, manufacturers shall use the HPDC Open Standard guidelines for Special Conditions when inventorying, screening and assessing these materials and substances in product reports and declarations.

Additional Guidance - Material Ingredient Screening and Optimization Action Plan (new for v4.1)

Project Teams: Compliant Material Ingredient Screening and Optimization Action Plans are standalone documents provided by manufacturers and are valued at ½ product for credit calculation purposes. Action plans are created after a manufacturer has inventoried their product to 1000 ppm and have provided a publicly available inventory and detailed action plan for how the product plans to mitigate hazard flagged ingredients.

Guidance for Manufacturers Creating Material Ingredient Screening & Optimization Action Plan Reports:

A compliant Action Plan is developed by a manufacturer and based upon analysis from conducting a product inventory of ingredients. The manufacturer has inventoried the product to at least 1,000 ppm and has conducted a screening or assessment to determine ingredient hazard scores/classes within each product via any of the material ingredient reporting formats in credit Option 1. For the Action Plan, the manufacturer has identified specific opportunities for improvement and has provided a detailed action plan to mitigate or reduce known hazards following the principles of green chemistry. The action plan is a publicly available report and must be product-specific (not company, manufacturer or brand), and include the required information below.

A compliant Action Plan is a standalone document that includes these elements:

- ▶ Unique document ID number of the report
- ▶ Product type
- ▶ Product name(s) and/or product line(s) covered by the report
- ▶ Date of creation of the action plan
- ▶ Expiration date for the report (a maximum of 5 years from the date of creation)

- ▶ Description of the inventory conducted including the dataset, software, and screening or assessment platform used by the manufacturer to complete the ingredient screening and analysis.
- ▶ Provide a link to the underlying publicly available inventory meeting the requirements of Option 1, if not available in the action plan report.
- ▶ A description of specific green chemistry principles targeted for implementation in the action plan. Include proposed changes in formulation or manufacturing processes that are planned as part of green chemistry optimization strategy.
- ▶ A table or otherwise summary of the specific green chemistry principles targeted for implementation in the action plan, including specific dates and a timeline for completion of all the steps described in the action plan.
- ▶ Contact information of the organizational representative responsible for implementation and success of the proposed action plan.
- ▶ Include a written narrative describing the analysis of product ingredients and the action plan steps identified to address specific improvement areas. Specifically, the action plan narrative must include a descriptions of immediate and long-term actions that will be pursued to reduce hazards within the product and which principle(s) of green chemistry are being adopted by the organization (per ingredient) to make the changes within the 5 year timeframe.
 - Examples of optimization strategies in an action plan include: identifying planned manufacturing or formulation changes; planned alternatives assessment; proposed changes in manufacturing processes; and describing how those changes are expected to achieve the end goal of safer chemistry for the product.
 - The principles of green chemistry are those created by Paul Anastas and John Warner, www.warnerbabcock.com/green-chemistry/the-12-principles
- ▶ An action plan complete with all the requirements stated above earns 0.5 product valuation.

A sample of a compliant Material Ingredient Action Plan is found in the Resources section of the online LEED credit library.

Additional Guidance - Advanced Inventory and Assessment Pathway (new for v4.1)

Project Team Members and Manufacturers: Compliant Advanced Inventory and Assessment reports are third-party verified standalone documents provided by manufacturers and are valued as 1 product for credit calculation purposes. The end use product has demonstrated a product inventory and assessment of ingredients using any of the following programs:

The following product certifications and reports are eligible:

- ▶ *Cradle to Cradle*: Product has Material Health Certificate or is Cradle to Cradle Certified™ under standard version 3 or later with a Material Health achievement level at the Bronze level or higher, value at 100% by Cost or 1 Product.
- ▶ *Declare*: Product has a Declare label that is third-party verified and “Red List Free” or “LBC Red List Free.”
- ▶ *Green Seal*. Products certified under the Standard for Paints, Coatings, Stains and Sealers (GS-11, Edition 4.0) that meet LEEDv4.1 Material Ingredients Credit: Option 2 – Material Ingredient Optimization criteria [i.e. they do not include GHS Reproductive toxins (categories 1 and 2)]. Qualifying products will include the following language on their certificates and in their Green Seal-certified product directory listings to indicate compliance with LEED v4.1 criteria:
 - “Eligible for LEEDv4.1 Material Ingredients Credit: Option 2 – Material Ingredient Optimization”
- ▶ *Health Product Declaration*: The product has demonstrated a chemical inventory to at least 0.01% by weight (100 ppm) with no GreenScreen LT-1 hazards or GHS Category 1 hazards. The HPD must be third party verified.
- ▶ *Health Product Declaration*: The product has demonstrated a chemical inventory to at least 0.01% by weight (100ppm) and at least 75% by weight of product is assessed using GreenScreen Benchmark assessment and/or is inventoried using applicable HPDC Special Conditions reporting requirements. The remaining 25% by weight of product has been

inventoried. The GreenScreen assessment must be publicly available. The HPD must be third-party verified.

- ▶ *Living Product Challenge*. Living Product Challenge certified products that include a “Red List Free” or “LBC Red List Free” Declare label.
- ▶ *Manufacturer Inventory*: The product has demonstrated a chemical inventory to at least 0.01% by weight (100 ppm) with no GreenScreen LT-1 hazards or GHS Category 1 hazards. The manufacturer inventory must be third party verified.
- ▶ *Manufacturer Inventory*: The product has demonstrated a chemical inventory to at least 0.01% by weight (100ppm) and at least 75% by weight of product is assessed using GreenScreen Benchmark assessment. The remaining 25% by weight of product has been inventoried. The GreenScreen assessment must be publicly available. The manufacturer inventory must be third-party verified.

Guidance for Manufacturers pursuing the Advanced Inventory and Assessment Pathway

To be eligible for showing an advanced inventory and assessment, a manufacturer has conducted a compliant inventory and/or assessment and published a document that satisfies the criteria below. All of the reports must be verified by an approved third-party verifier/assessor meeting the third-party verification requirements in section “*Third-Party Verification Program Requirements for Material Ingredient Reporting and Optimization*.”

- ▶ Cradle to Cradle Bronze Certification or Material Health Certificate at a Bronze level
 - Product has Material Health Certificate or is Cradle to Cradle Certified™ under standard version 3 or later with a Material Health achievement level at the Bronze level. Cradle to Cradle product certificates meet the third-party verification criteria.
- ▶ Declare: Third-party verified “Red List Free” or “LBC Red List Free” Declare labels or Living Product Challenge labels.
 - The Declare label must indicate third-party verification.

Health Product Declaration or Manufacturer Inventory (with no GreenScreen List Translator-1 hazards):

- The product must demonstrate a chemical inventory to at least 0.01% by weight (100 ppm) with no GreenScreen LT-1 hazard scores or GHS Category 1 hazards.
- Since this pathway is an intermediate step towards full optimization and places greater emphasis on advanced inventory and assessment rather than major substitution or elimination of chemicals, GS scores of List Translator- probable carcinogens (LT-P1) and List Translator- Unknown (LT-UNK) are allowed to be reported as is for the ingredients and do not need to be resolved further by the manufacturer in order for this document to be compliant with LEED v4.1 requirements.
- Products utilizing the GHS pathway must display no Category 1 hazards for each ingredient down to 100 ppm levels. If there is no endpoint for a chemical, the manufacturer will need to display “no hazard endpoint” in their manufacturer inventory disclosure document. This reporting must be different from reporting of a data gap, if there is insufficient data available for a particular endpoint, the manufacturer will need to report “Insufficient data” for the particular endpoint of that ingredient/chemical.
- Chemicals with form specific hazards or special conditions must follow the special conditions guidelines in manufacturer inventory/HPDs, these must be available on HPDC website.
- The HPD or Manufacturer Inventory must be third party verified by an approved third-party verifier/assessor via either HPDC or Clean Production Action and meet the third-party verification requirements as specified above.
- In addition to the requirements above, compliant manufacturer inventories must follow the most recent version of the HPDC Open Standard for reporting, ingredient categorization, disclosure thresholds, displaying content ranges, hazards, and other special conditions.
- ▶ Health Product Declaration or Manufacturer Inventory (with 75% by weight of product assessed):

- At least 75% by weight of the product to 100ppm level for ingredients must be assessed using the GreenScreen Benchmark methodology.
- Since this pathway is an intermediate step towards full optimization and places greater emphasis on advanced inventory and assessment rather than major substitution or elimination of chemicals, the document must show that 75% by weight of the chemistry within the overall product has been assessed using a full GreenScreen Benchmark Assessment methodology. For the remaining 25% by weight of the product, GreenScreen List Translator scores of ingredients (i.e. preliminary GS List Translator screening) associated with their chemistry has been conducted.
- GreenScreen assessment reviews must be conducted via an approved GreenScreen Profiler and the Health Product Declaration or Manufacturer Inventory must be third-party verified by an approved verifier that meets the third-party verification requirements as stated above.
- Any GreenScreen assessments that resolve unknown and/or probable hazard scores must be made available to the public or the industry, either online in a free directory or through subscription to a common industry database.
- In addition to the requirements above, compliant manufacturer inventories must follow the most recent version of the HPDC Open Standard for reporting, ingredient categorization, disclosure thresholds, displaying content ranges, hazards, and other special conditions.

All compliant reports must also include:

- ▶ Product name(s) and/or product line(s) covered by the report
- ▶ Unique document ID number of the report
- ▶ Expiration date for the report

Additional Guidance – Material Ingredient Optimization Pathway (new for v4.1)

Project Team Members and Manufacturers: Compliant Material Ingredient Optimization reports are third-party verified standalone documents provided by manufacturers and valued at 1.5 products for credit calculation purposes. The end use product has demonstrated a product inventory and assessment of ingredients using any of the following programs:

- ▶ *Cradle to Cradle*. Product is Cradle to Cradle v3 (or later) certified with Material Health category score of Silver or higher, or a Cradle to Cradle certified Material Health Certificate at Silver level or higher.
- ▶ *Health Product Declaration*: The product has demonstrated a chemical inventory to at least 0.01% by weight (100ppm) and at least 95% by weight of product is assessed using GreenScreen Benchmark assessment and/or is inventoried using applicable HPDC Special Conditions reporting requirements. No Benchmark 1 hazards (BM-1) are present in the end use product. The remaining 5% by weight of product not assessed has been inventoried and screened using GreenScreen List Translator and no GreenScreen LT-1 hazards are present in the end use product. The documents must be third party verified.
- ▶ *Living Product Challenge*. Products certified to the Living Product Challenge which includes achievement of Imperative 09: Transparent Material Health.
- ▶ *Manufacturer Inventory*: The product has demonstrated a chemical inventory to at least 0.01% by weight (100ppm) and at least 95% by weight of product is assessed using GreenScreen Benchmark assessment. No Benchmark 1 hazards (BM-1) are present in the end use product. The remaining 5% by weight of product not assessed has been inventoried and screened using GreenScreen List Translator and no GreenScreen LT-1 hazards are present in the end use product. The documents must be third party verified.

Guidance for Manufacturers pursuing the Material Ingredient Optimization Pathway

To be eligible for showing a Material Ingredient Optimization, a manufacturer has conducted a compliant inventory and/or assessment and published a document that satisfies the criteria below. All

of the reports must be verified by an approved third-party verifier/assessor meeting the third-party verification requirements in section “*Third-Party Verification Program Requirements for Material Ingredient Reporting and Optimization.*”

- ▶ Cradle to Cradle Silver or higher Certification or Material Health Certificate at a Silver level
 - Product has Material Health Certificate or is Cradle to Cradle Certified™ under standard version 3 or later with a Material Health achievement level at the Silver level.
- ▶ Health Product Declaration or Manufacturer Inventory (with 95% by weight of product assessed)
 - At least 95% by weight of the product to the 100ppm level for constituents must be assessed using the GreenScreen Benchmark methodology.
 - Since this pathway requires full optimization in the form of major substitution or elimination of most hazardous chemicals in the product, the document will display that at least 95% by weight of the chemistry within the overall product has been assessed against full GreenScreen Benchmark Assessment methodology (with no BM-1 hazards in end use products) as opposed to just List Translator score hazards (LT-1, LT-P1, LT-UNK). The remaining 5% by weight of product not assessed has been inventoried and screened using GreenScreen List Translator and no GreenScreen LT-1 hazards are present in the end use product.
 - GreenScreen assessments must be conducted via an approved GreenScreen Profiler and the Health Product Declaration or Manufacturer Inventory must be third-party verified.
- ▶ In addition to the requirements above, compliant manufacturer inventories must follow the most recent version of the HPDC Open Standard for reporting, ingredient categorization, disclosure thresholds, displaying content ranges, hazards, and other special conditions. Living Product Challenge.
 - Products certified to the Living Product Challenge which includes achievement of Imperative 09: Transparent Material Health.

All compliant reports must also include:

- ▶ Product name(s) and/or product line(s) covered by the report
- ▶ Unique document ID number of the report
- ▶ Expiration date for the report

International Tips – Material Ingredient Optimization (Option 2)

Project Team Members and Manufacturers: Follow the LEED v4 Reference Guide for projects using the REACH Optimization ACP with the following additional guidance.

The following programs and documents are eligible for the international ACP. Products are valued at 1 product.

- ▶ *REACH Optimization.*
 - End use products and materials have fully inventoried chemical ingredients and each substance has been screened against the Authorization List – Annex XIV, the Restriction list – Annex XVII and the SVHC candidate lists:
 - *Restriction List:* <https://echa.europa.eu/substances-restricted-under-reach>
 - *Authorization list:* <https://echa.europa.eu/authorisation-list>
 - *SVHC Candidate list:* <https://echa.europa.eu/candidate-list-table>
 - Products must not contain any intentionally added ingredients found on the REACH Restricted and Authorization lists. Any ingredients in the product found on the Candidate list are not included at concentrations above 100ppm.
- ▶ *Global Green Tag.* Product has a certified Product Health Declaration (PhD) report.

Guidance for Manufacturers Providing Documentation for the International ACP:

REACH Optimization: Manufacturers shall provide documentation that confirms a product has had a full ingredient inventory of all intentionally added substances found in the end use product and that a REACH assessment has been conducted. The REACH compliance documentation shall state that no intentionally added ingredients are present in the product that are listed on the REACH Restriction and Authorization lists. Further, the manufacturer documentation shall verify that any ingredients found on the REACH Candidate list are not present in concentrations above 100ppm.

Acceptable documentation:

- Transparency report: Standalone transparency report that provides a complete inventory and shows compliance with the *REACH International ACP* credit requirements. Documentation must include a unique document ID number for the report, list the product type, product name(s) and/or product line(s) covered by the report, date of creation of the report, and expiration date for the report (a maximum of 5 years from the date of creation).

OR

- Manufacturer Attestation: Standalone letter from the product manufacturer declaring that the product meets the *REACH International ACP* credit requirements. No inventory is necessary. This letter must include a signature from someone at the company knowledgeable of product formulations and include their name, job title, and date of signature. The letter must also include the product type, product name(s) and/or product line(s) covered by the attestation letter. Manufacturer letters shall expire when product formulations change, or 3 years from the date of letter signature.

Global Green Tag: Manufacturers have a certified Global Green Tag Product Health Declaration (PHD) and the document is available publicly.

All compliant reports must also include:

- ▶ Product name(s) and/or product line(s) covered by the report
- ▶ Product type and product description
- ▶ Unique document ID number of the report
- ▶ Expiration date for the report

Third-Party Verification Program Requirements for Material Ingredient Reporting and Optimization

- ▶ Qualifying third-party verification and certification programs should conduct their operations in compliance with ISO 17065 or have been certified to ISO 17065.
- ▶ The verifier will confirm that the output document was conducted in compliance with the requirements of the LEED credit.
- ▶ The verification process must cover an additional step of verifying the reliability of content inventory of the product as well as verifying product compliance to required thresholds in LEED.
- ▶ The third-party verifier of the material ingredient documents must be independent from the individual that created the documents.
- ▶ The verifier must be qualified to conduct verification of material ingredient documents.
- ▶ The material ingredient reporting program must provide training for the verifier on the verification process and updates to the program regularly to ensure consistency of verification.
- ▶ The program must have a database of publicly available third-party verified products that meet the applicable LEED credit requirements.
- ▶ The program must have a written third-party verification process and procedure that is updated regularly with a process to implement updates and changes to verifiers of the program that is publicly available.

Required Documentation

Follow LEED v4 reference guide documentation requirements with the following modifications:

- ▶ Option 1: use the LEED v4.1 MR building product calculator. For products that have been verified for LEED by GBCI, include “Verified for LEED Documentation” ID numbers, or submit copies of product inventory documents in LEED Online for products contributing toward credit.
- ▶ Option 2: use the LEED v4.1 MR building product calculator. For products that have been verified for LEED by GBCI, include “Verified for LEED Documentation” ID numbers for products selected in the calculator, or copies of action plans and optimization reports in LEED Online for products contributing toward credit.
- ▶ Option 3: Documentation requirements no longer applicable.

Note: All product reports must be valid at the time the product was purchased for the project.

Exemplary Performance

Option 1: Purchase at least 40 qualifying permanently installed building products from ten different manufacturers that meet the credit criteria (Source at least 20 products from five different manufacturers for CS and Warehouse & Distribution Centers).

Option 2: Purchase at least 10 qualifying products from five different manufacturers of permanently installed building products that meet the credit criteria.

Connection to Ongoing Performance

- ▶ LEED O+M MR credit Purchasing: products meeting the above requirements can help achieve the MR credit Purchasing .

MR Credit: PBT Source Reduction – Mercury

This credit applies to

- ▶ BD+C: Healthcare (1 point)

INTENT

To reduce the release of persistent, bioaccumulative, and toxic (PBTs) chemicals associated with the life cycle of building materials.

REQUIREMENTS

HEALTHCARE

Specify and install fluorescent lamps with both low mercury content (MR Prerequisite PBT Source Reduction—Mercury) and long lamp life, as listed in Table 1.

Table 1. Criteria for rated life of low-mercury lamps

<i>Lamp</i>	<i>Maximum content</i>	<i>Lamp life (hrs)</i>
T-8 fluorescent, eight-foot	10 mg mercury	Standard output - 24,000 rated hours on instant start ballasts (3-hour starts) High output - 18,000 rated hours on instant start ballasts or program start ballasts (3-hour starts)
T-8 fluorescent, four-foot	3.5 mg mercury	Both standard and high output - 30,000 rated hours on instant start ballasts, or 36,000 rated hours on program start ballasts (3 hour starts)
T-8 fluorescent, two-foot and three-foot	3.5 mg mercury	24,000 rated hours on instant start ballasts or program start ballasts (3-hour starts)
T-8 fluorescent, U-bent	6 mg mercury	18,000 rated hours on instant start ballasts, or 24,000 rated hours on program start ballasts (3-hour starts)
T-5 fluorescent, linear	2.5 mg mercury	Both standard and high-output - 25,000 rated hours on program start ballasts
T-5 fluorescent, circular	9 mg mercury	Both standard and high-output - 25,000 rated hours on program start ballasts
Compact fluorescent, nonintegral ballast	3.5 mg mercury	12,000 rated hours
Compact florescent, integral ballast, bare bulb	3.5 mg mercury, ENERGY STAR qualified	Bare bulb - 10,000 rated hours Covered models such as globes, reflectors, A-19s - 8,000 hours
High-pressure sodium, up to 400 watts	10 mg mercury	Use noncycling type or replace with LED lamps or induction lamps
High-pressure sodium, above 400 watts	32 mg mercury	Use noncycling type or replace with LED lamps or induction lamps

Do not specify or install circular fluorescent lamps or probe start metal halide lamps.

GUIDANCE

There are no substantive changes to the credit requirements; refer to the LEED v4 reference guide, with the following addition:

Further Explanation**Connection to Ongoing Performance**

- ▶ LEED O+M MR credit Purchasing: This credit is a strategy that can help achieve the MR credit Purchasing in O+M v4.1 rating system.

MR Credit: PBT Source Reduction – Lead, Cadmium, and Copper

This credit applies to

- ▶ BD+C: Healthcare (1-2 points)

INTENT

To reduce the release of persistent, bioaccumulative, and toxic (PBT) chemicals associated with the life cycle of building materials.

REQUIREMENTS

HEALTHCARE

Specify substitutes for materials manufactured with lead and cadmium, as follows.

Lead

- ▶ For water intended for human consumption, specify and use solder and flux to connect plumbing pipe on site that meets the California AB1953 standard, which specifies that solder not contain more than 0.2% lead, and flux not more than a weighted average of 0.25% lead for wetted surfaces. The “lead free” label as defined by the Safe Drinking Water Act (SDWA)) does not provide adequate screening for the purposes of this credit because the SDWA defines “lead free” as solders and flux containing 0.2% lead or less.
- ▶ For water intended for human consumption, specify and use pipes, pipe fittings, plumbing fittings, and faucets that meet the California law AB1953 of a weighted average lead content of the wetted surface area of not more than 0.25% lead.
- ▶ Specify and use lead-free roofing and flashing.
- ▶ Specify and use electrical wire and cable with lead content less than 300 parts per million.
- ▶ Specify no use of interior or exterior paints containing lead.
- ▶ For renovation projects, ensure the removal and appropriate disposal of disconnected wires with lead stabilizers, consistent with the 2002 National Electric Code requirements.

Lead used for radiation shielding and copper used for MRI shielding are exempt.

Cadmium

- ▶ Specify no use of interior or exterior paints containing intentionally added cadmium.

Copper

- ▶ For copper pipe applications, reduce or eliminate joint-related sources of copper corrosion:
 - use mechanically crimped copper joint systems; or
 - specify that all solder joints comply with ASTM B828 2002, and specify and use ASTM B813 2010 for flux.

GUIDANCE

There are no substantive changes to the credit requirements; refer to the LEED v4 reference guide, with the following addition:

Further Explanation

Connection to Ongoing Performance

- ▶ LEED O+M MR credit Purchasing: This credit is a strategy that can help achieve the MR credit Purchasing in O+M v4.1 rating system.

MR Credit: Furniture and Medical Furnishings

This credit applies to

- ▶ BD+C: Healthcare (1-2 points)

INTENT

To enhance the environmental and human health performance attributes associated with freestanding furniture and medical furnishings.

REQUIREMENTS

HEALTHCARE

Use at least 30% (1 point) or 40% (2 points), by cost, of all freestanding furniture and medical furnishings (e.g., mattresses, foams, panel fabrics, cubicle curtains, window coverings, other textiles) that meet the criteria in one of the following three options.

Include built-in casework and built-in millwork in the base building calculations, even if manufactured off site. The dollar value of any individual product may be included in the total qualifying value if the product meets the criteria.

Option 1. Minimal Chemical Content

All components that constitute at least 5%, by weight, of a furniture or medical furnishing assembly, including textiles, finishes, and dyes, must contain less than 100 parts per million (ppm) of at least four of the five following chemical groups:

- ▶ urea formaldehyde;
- ▶ heavy metals, including mercury, cadmium, lead, and antimony;
- ▶ hexavalent chromium in plated finishes consistent with the European Union Directive on the Restriction of the Use of Certain Hazardous Substances (EU RoHS);
- ▶ stain and nonstick treatments derived from perfluorinated compounds (PFCs), including perfluorooctanoic acid (PFOA); and
- ▶ added antimicrobial treatments.

AND/OR

Option 2. Testing and Modeling of Chemical Content

All components of a furniture or medical furnishing assembly, including textiles, finishes, and dyes, must contain less than 100 parts per million (ppm) of at least two of the five chemicals or materials listed in Option 1.

New furniture or medical furnishing assemblies must be in accordance with ANSI/BIFMA Standard Method M7.1-2011. Comply with ANSI/BIFMA e3-2019 Furniture Sustainability Standard, Sections 7.6.1 and 7.6.2, using either the concentration modeling approach or the emissions factor approach. Model the test results using the open plan, private office, or seating scenario in ANSI/BIFMA M7.1, as appropriate. USGBC-approved equivalent testing methodologies and contaminant thresholds are also acceptable. Documentation submitted for furniture must indicate the modeling scenario used to determine compliance.

Salvaged and reused furniture more than one year old at the time of use is considered compliant, provided it meets the requirements for any site-applied paints, coatings, adhesives, and sealants.

AND/OR

Option 3: Building Product Disclosure and Optimization

Use products that meet at least one of the criteria below. Each product can receive credit for each criterion met. The scope of any environmental product declaration (EPD) must be at least cradle to gate.

- ▶ Life-cycle assessment and environmental product declarations.
 - Products with a publicly available, critically reviewed life-cycle assessment conforming to ISO 14044 that have at least a cradle to gate scope are valued as one whole product for the purposes of credit achievement calculation.
 - Product-specific Type III EPD -- Internally Reviewed. Products with an internally critically reviewed LCA in accordance with ISO 14071. Products with product-specific internal EPDs which conform to ISO 14025, and EN 15804 or ISO 21930 and have at least a cradle to gate scope are valued as one whole product for the purposes of credit achievement calculation.
 - Industry-wide Type III EPD -- Products with third-party certification (Type III), including external verification, in which the manufacturer is explicitly recognized as a participant by the program operator. Products with industry-wide EPDs, which conform to ISO 14025, and EN 15804 or ISO 21930 and have at least a cradle to gate scope are valued as one whole product for the purposes of credit achievement calculation.
- ▶ Environmental Product Declarations which conform to ISO 14025, 14040, 14044, and EN 15804 or ISO 21930 and have at least a cradle to gate scope.
 - Product-specific Type III EPD -- Products with third-party certification (Type III), including external verification and external critical review are valued as 1.5 products for the purposes of credit achievement calculation.
- ▶ *Extended producer responsibility.* Products purchased from a manufacturer (producer) that participates in an extended producer responsibility program or is directly responsible for extended producer responsibility. Products meeting extended producer responsibility criteria are valued at 50% of their cost for the purposes of credit achievement calculation.
- ▶ *Bio-based materials.* Bio-based products and materials other than wood must be tested using ASTM Test Method D6866 or equivalent method ISO 16620-2, or be certified to the USDA BioPreferred Voluntary Labeling Initiative that includes verification via ASTM 6866 testing. Exclude hide products, such as leather and other animal skin material.
 - Bio-based products that meet the criteria above: value at 50% of cost multiplied by the biobased content of the product for the purposes of credit achievement calculation.
- ▶ *Wood products.* Wood products must be certified by the Forest Stewardship Council or USGBC-approved equivalent. Products meeting wood products criteria are valued at 100% of their cost for the purposes of credit achievement calculation.
- ▶ *Materials reuse.* Reuse includes salvaged, refurbished, or reused products. Products meeting materials reuse criteria are valued at 200% of their cost for the purposes of credit achievement calculation.
- ▶ *Recycled content.* Products meeting recycled content criteria are valued at 100% of their cost for the purposes of credit achievement calculation.
 - Recycled content is the sum of postconsumer recycled content plus one-half the preconsumer recycled content, based on weight.
 - The recycled fraction of the assembly is then multiplied by the cost of assembly to determine the recycled content value.

Products that meet the above criteria are valued according to source location (extraction,

manufacture, and purchase point must be within the distances noted below):

For credit achievement calculation, products sourced (extracted, manufactured, purchased) within 100 miles (160 km) of the project site are valued at 200% of their base contributing cost.

GUIDANCE

There are no substantive changes to the credit requirements; refer to the LEED v4 reference guide, with the following additions:

Further Explanation

Option 3: Building Product Disclosure and Optimization

Please refer to the v4.1 beta guide section for MR credits Environmental Product Declarations and Sourcing of Raw Materials for guidance on implementation.

Connection to Ongoing Performance

- ▶ LEED O+M MR credit Purchasing: This credit is a strategy that can help achieve the MR credit Purchasing in O+M v4.1 rating system.

MR Credit: Design for Flexibility

This credit applies to

- ▶ BD+C: Healthcare (1 point)

INTENT

Conserve resources associated with the construction and management of buildings by designing for flexibility and ease of future adaptation and for the service life of components and assemblies.

REQUIREMENTS

HEALTHCARE

Increase building flexibility and ease of adaptive use over the life of the structure by employing at least three of the following strategies.

- ▶ Use *interstitial space*. Design distribution zone utility systems and equipment including HVAC, plumbing, electrical, information technology, medical gases, and life safety systems to serve the occupied zones and have the capacity to control multiple zones in clinical spaces.
- ▶ Provide programmed soft space, such as administration or storage, equal to at least 5% of *departmental gross area* (DGA). Locate soft space adjacent to clinical departments that anticipate growth. Determine a strategy for future accommodation of displaced soft space.
- ▶ Provide shell space equal to at least 5% of DGA. Locate it such that it can be occupied without displacing occupied space.
- ▶ Identify horizontal expansion capacity for diagnostic and treatment or other clinical space equal to at least 30% of existing floor area (excluding inpatient units) without demolition of occupied space (other than at the connection point). Reconfiguration of additional existing occupied space that has been constructed with demountable partition systems is permitted.
- ▶ Design for future vertical expansion on at least 75% of the roof, ensuring that existing operations and service systems can continue at or near capacity during the expansion.
- ▶ Designate space for future above-grade parking structures equal to 50% of existing on-grade parking capacity, with direct access to the main hospital lobby or circulation. Vertical transportation pathways that lead directly to the main hospital lobby or circulation are acceptable.
- ▶ Use demountable partitions for 50% of applicable areas.
- ▶ Use movable or modular casework for at least 50% of casework and custom millwork. Base the calculation on the combined value of casework and millwork, as determined by the cost estimator or contractor.

GUIDANCE

There are no substantive changes to the credit requirements; refer to the LEED v4 reference guide, with the following addition:

Further Explanation

Connection to Ongoing Performance

- ▶ LEED O+M MR prerequisite Waste Performance: This credit is a strategy that can help achieve the MR prerequisite Waste Performance in O+M v4.1 rating system.

MR Credit: Construction and Demolition Waste Management

This credit applies to

- ▶ BD+C: New Construction (1-2 points)
- ▶ BD+C: Core & Shell (1-2 points)
- ▶ BD+C: Schools (1-2 points)
- ▶ BD+C: Retail (1-2 points)
- ▶ BD+C: Data Centers (1-2 points)
- ▶ BD+C: Warehouses & Distribution Centers (1-2 points)
- ▶ BD+C: Hospitality (1-2 points)
- ▶ BD+C: Healthcare (1-2 points)

INTENT

To reduce construction and demolition waste disposed of in landfills and incineration facilities through waste prevention and by reusing, recovering, and recycling materials, and conserving resources for future generations. To delay the need for new landfill facilities that are often located in frontline communities and create green jobs and materials markets for building construction services.

REQUIREMENTS

NC, CS, SCHOOLS, RETAIL NC, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY NC, HEALTHCARE

Develop and implement a construction and demolition waste management plan and achieve points through waste prevention and/or diversion.

Waste Management Plan and Report:

All projects must develop and implement a construction and demolition waste management plan:

- Identify strategies to reduce the generation of waste during project design and construction.
- Establish waste diversion goals for the project by identifying the materials (both structural and nonstructural) targeted for diversion.
- Describe the diversion strategies planned for the project. Describe where materials will be taken including expected diversion rates for each material.

Provide a final waste management report detailing all waste generated, including disposal and diversion rates for the project. Calculations can be by weight or volume but must be consistent throughout. Exclude excavated soil and land-clearing debris from calculations. Include materials destined for alternative daily cover (ADC) in the calculations as waste (not diversion). Any materials sent to a comingled recycling facility for processing must take the facility average recycling rate and must include any ADC as waste (not diversion).

Option 1. Diversion (1 point)

Follow the Waste Management Plan and divert at least 50% of the total construction and demolition materials from landfills and incineration facilities.

AND/OR

Option 2. Waste Prevention (2 points)

Follow the Waste Management Plan and prevent waste through reuse and source reduction design strategies. Salvage or recycle at least 50% of demolition debris and utilize waste minimizing design strategies and construction techniques for new construction elements. Track all demolition debris generated by the project from start of construction through project completion. Also track all new construction waste materials generated from start of construction through project completion to determine the project's total waste generation from new construction activities. Exclude hazardous materials and land-clearing debris from calculations.

Points are awarded as follows:

Divert at least 50% of all demolition waste, if any. Generate less than 10 lbs./ft² (50 kg/m²) of waste materials from all new construction activities (2 points)

GUIDANCE

Refer to the LEED v4 reference guide, with the following additions and modifications:

Behind the Intent

Beta Update

The requirement for diversion via multiple material streams in LEED has been removed. Project teams now have more flexible ways to prevent and divert waste in order to earn points. This new structure rewards the hierarchy of first reduce, reuse, and then recycle.

Additional updates clarify when waste-to-energy can count as diversion for international projects. The changes also incorporate and modify the popular *Pilot Credit 87 for Verified Construction & Demolition Recycling Rates* into the credit (via Exemplary Performance) and define the minimum requirements for a commingled facility certified recycling rate. Finally, the total waste generation pathway has been revised to favor project types that source reduce and generate less waste overall.

Step-by-Step Guidance

Refer to LEED v4 Construction and Waste Management Plan prerequisite reference guide with the following modifications:

All recycling facilities must be regulated by a local or state authority. Note that regulatory authorities often do not regulate recycling rates of facilities. To determine the mixed waste processing facility recycling rate (also known as “commingled” or “mixed materials” recycling rates), projects must use an average diversion rate for the facility that corresponds to the time materials were generated on the project and sent to the facility. The average recycling rate for the facility must exclude ADC.

Construction Waste Management Plan

Step 1: Develop and implement a Construction Waste Management Plan during the project design phase and prior to construction.

All projects must develop a Construction Waste Management Plan per the LEED v4 Reference guide with modifications made in the credit requirements. The Waste Management Plan should include goals for waste prevention, waste diversion, identification of recycling haulers and recycling facilities, and data collection and reporting procedures.

- Tracking: develop a method for tracking the amount of all waste and recyclable materials generated during demolition activities (if any), including demolition of existing buildings, other structures on the site, or components demolished as part of building renovation work.

Separately, keep track of waste and recyclable materials generated during new construction activities.

- ▶ Waste prevention: Include strategies targeted to reduce the total amount of waste generated during construction, renovation or demolition activities. Identify design strategies to dematerialize or otherwise prevent unused or waste materials. Deconstruct and salvage renovation materials for reuse to the greatest extent possible.
- ▶ Diversion: locate recyclers or organizations that provide recycling options for the materials targeted for diversion. Confirm each facility can accept the types of materials the project plans to send for recycling. Obtain a diversion rate for the facility/organization receiving materials, including ADC amounts for commingled recycling processors. Indicate if any commingled facilities have third party verification of recycling rates. Projects must take the facility average recycling rate and be reported in weight or volume (must be consistent). Volume to weight conversions are acceptable if conversion factors are developed by reputable local agency or organizations.
 - If no local values are available, refer to the most recent EPA Volume-to-Weight Conversion Factors for Solid Waste: www.epa.gov/smm/volume-weight-conversion-factors-solid-waste

Indicate in the Construction Waste Management Plan whether or not the selected recycling facilities that process commingled (mixed) materials have third party verification of their recycling rates. These facilities support accurate reporting and nationally consistent metrics for recycling rates. See the *Exemplary Performance* section for minimum criteria for eligible third-party verified recycling rate programs.

Step 2 – Implement the Waste Management Plan: Follow the LEED v4 Reference Guide for step-by-step guidance (Step #1) in the Construction Waste Management Credit. Note that project-specific waste diversion percentages from waste sorting facilities are not allowed. Projects must use the facility's average recycling rate.

Step 3 – Calculate Diversion Rate: Follow the LEED v4 Reference Guide for step-by-step guidance (Step #2) in the Construction Waste Management Credit. Note that the number of material streams are no longer required to be tracked. Develop a method for tracking the amount of all demolition materials generated on the project (if any). Separately, keep track of waste materials generated during new construction activities.

Step 4 – Produce Waste Management Report: Follow the LEED v4 Reference Guide for step-by-step guidance (Step #3) in the Construction Waste Management Credit with modifications for the v4.1 changes herein.

Option 1: Diversion

After exploring source reduction and design strategies to prevent waste, determine strategies for on-site and off-site waste collection during construction and consider the infrastructure needed for implementation. Projects may use a combination of on-site separation and commingled collection to achieve the diversion goals, depending on what is appropriate for the project location, waste materials generated, and available facilities and haulers. Strategies for achieving high waste diversion include:

- ▶ Stage collection bins onsite to correspond with construction phases and contractor schedules. If one trade is onsite for a defined period that has a recyclable waste stream, consider having a single bin for that type of waste instead of—or in addition to—a commingled bin (examples include a bin for concrete recycling during demolition, or separate bins for drywall, wood framing, or roofing waste during those phases).

- ▶ Source separated materials taken to an individual recycler (such as a scrap metal recycler, sheetrock vendor, or biomass operator) will tend to have much higher diversion rates than a mixed-waste recycling processor.
- ▶ Donate surplus or salvaged materials.
- ▶ Participate in manufacturer take-back and recycling programs for removed products in salvageable condition, like ceiling tiles, furniture or flooring.
- ▶ Incineration of some C&D materials (other than wood) may be considered diversion for international projects only if reuse and recycling methods are not readily available in the project's location; this must be included in the Waste Management Plan. See *Further Explanation* for additional details on waste-to-energy.

Option 2: Waste Prevention:

Follow step-by-step instructions from the v4 Reference Guide with the following modifications:

Step 2: Calculate Total Waste Reduction

Projects can document waste prevention techniques undertaken on the project to earn 2 points. To calculate the project's total waste, include all materials generated from the project construction and demolition activities and sent offsite for recycling, diversion, landfill, incineration, or any other uses. Exclude hazardous materials, land-clearing debris, and on-site reused materials from the generation numbers. A narrative, provided in the LEED form, must describe strategies in design and construction that were implemented to reduce waste from being generated on the jobsite (See LEED v4 reference guide, *Further Explanation*, *Source Reduction* for more guidance on source reduction strategies).

Tracking: To calculate waste prevention, project teams must track the amount of waste generated during demolition work and—separately—during new construction activities. For projects with demolition activities, including demolition of existing buildings, other structures on the site, or components demolished as part of building renovation work, develop a method for tracking the amount of all waste generated from demolition. To be eligible for Option 2, the project must demonstrate that at least 50% of the demolition waste was diverted (note: ADC does not count as diversion). Separately, keep track of waste generated during new construction activities. The amount of new construction-phase waste will be used to quantify the amount of waste generated (per ft² or m²) for Option 2 credit compliance.

Reducing waste by thoughtful design results in the reduction of waste on the jobsite. Stopping waste before it is created is always higher priority than managing waste after construction. Successful projects have implemented the following strategies to prevent construction and demolition waste in LEED:

- ▶ Reuse buildings or building components. Materials that are reused onsite are excluded from the waste generation calculations. Sending materials offsite to salvage, donation or recycling, such as through a documented manufacturer take-back program, does not count as waste prevention but can count towards project diversion.
- ▶ Source reduction strategies should be incorporated into the design of the project and outlined in the Waste Management Plan. These strategies include reusing existing materials and components, designing for modular construction sizes and techniques, specifying reduced packaging from vendors, designing for industry-standard measurements, eliminating unnecessary finishes, and off-site prefabrication of components or assemblies.
- ▶ Work with subcontractors and/or finish material suppliers to eliminate or minimize packaging waste and reduce overall materials rather than send them for disposal or recycling.

Further Explanation

Waste-to-Energy

Refer to LEED v4 reference guide with the following modifications:

The combustion of wood materials resulting from recycling processing (also known as “wood-derived fuel” or “biomass”) is classified as an acceptable means of diversion for projects both in the US and internationally and is not considered waste-to-energy for LEED project diversion reporting purposes.

Forms of waste-to-energy (other than wood) are not widely utilized for construction and demolition waste management in the United States and it is not considered an acceptable means of diversion for projects within the US. For projects outside the US where waste-to-energy markets for construction and demolition materials may be more common, projects may be eligible to count waste-to-energy as diversion under the *Alternative Compliance Path for International Projects* (see *International Tips* section).

International Tips

Alternative Compliance Path for International Projects: Waste-to-Energy

International projects that cannot meet diversion requirements via reuse and recycling methods may consider waste-to-energy as diversion if the European Commission Waste Framework Directive 2008/98/EC and Waste Incineration Directive 2000/76/EC are followed. Furthermore, waste-to-energy facilities must meet applicable European Committee for Standardization (CEN) EN 303 standards.

- ▶ EN 303-1—1999/A1—2003, Heating boilers with forced draught burners
- ▶ EN 303-2—1998/A1—2003, Heating boilers with forced draught burners
- ▶ EN 303-3—1998/AC—2006, Gas-fired central heating boilers
- ▶ EN 303-4—1999, Heating boilers with forced draught burners
- ▶ EN 303-5—2012, Heating boilers for solid fuels
- ▶ EN 303-6—2000, Heating boilers with forced draught burners
- ▶ EN 303-7—2006, Gas-fired central heating boilers equipped with a forced draught burner

Project teams pursuing this compliance option must demonstrate that reuse and recycling strategies were exhausted before sending material to waste-to-energy facilities.

Required Documentation

- ▶ A compliant Construction and Demolition Waste Management Plan.
- ▶ A narrative for waste prevention/design strategies used on the project to achieve the waste threshold and calculation of total waste per area (if pursuing Waste Prevention points).
- ▶ MR construction and demolition waste management calculator that demonstrates total and diverted waste amounts and documentation of recycling rates for commingled facilities.
- ▶ International projects: justification narrative for use of WTE strategy for international teams and documentation of WTE facilities adhering to EN standards (if applicable)

Exemplary Performance

Achieve Option 1 and Option 2.

OR

Divert 50% or more of all waste and utilize certified recycling facilities for all commingled waste.

Certified Recycling Facilities

Qualified third-party organizations who certify facility average recycling rates include these minimum program requirements:

Requirements for Certified Recycling Facilities:

Projects must utilize a recycling facility that processes and recycles commingled (mixed) construction and demolition waste materials that has received independent third-party certification of their recycling rates. Qualified third-party organizations who certify facility average recycling rates include these minimum program requirements:

- ▶ The certification organization follows guidelines for environmental claims and third-party oversight, including ISO/IEC 17065:2012 and relevant portions of the ISO 14000 family of standards.
- ▶ The certification organization is an independent third party who continuously monitors "certified" facilities to ensure that the facilities are operating legally and meeting the minimum program requirements for facility certification and recycling rates.
- ▶ Certification organizations shall certify to a protocol that was developed on a consensus basis for recycling facility diversion rates that is not in a draft or pilot program.
- ▶ The methodology for calculating facility recycling rates must be:
 - Developed with construction and demolition recycling industry stakeholders and be specific to the construction and demolition recycling industry;
 - Must include a methodology that is applicable across broad regions (i.e. nationally); and,
 - A published and publicly available standard.
- ▶ Data submitted by the facilities to the certification organization in support of the recycling rate is audited. The audit includes, at a minimum: the evaluation of recyclable sales records, verification of facility sales into commodity markets, an assessment of downstream materials and how these materials are managed after they leave the site, monitoring off-site movement of materials, and a review of the facilities' customers weight tags information.
- ▶ Facilities submit data to the certification organization that supports the recycling rate, such as a mass balance recycling rate (tons in/tons out) for a twelve month period, or quarterly sorts completed and verified by an independent third party entity.
- ▶ Breakdown of materials (by type and by weight), including analysis of supporting data relating to amounts (in tons) and types of materials received and processed at the facility.
- ▶ At a minimum, the third-party certifying organization conducts an on-site visit of the Facility for the first year certification, with subsequent site visits occurring at least once every two (2) years, unless additional visits are deemed necessary by the certification organization. The site visit will examine:
 - How materials enter, are measured, deposited, processed/sorted and exit facility,
 - Conduct interviews with key personnel, and discuss how materials are managed after they leave the site
 - Confirm equipment types and capacity,
 - Observe and verify load/materials sorting and accuracy,
 - Verify use and accuracy of scales including calibration frequency.
- ▶ Diversion rates shall adhere to these requirements:
 - Measurements must be based on weight (not volume), using scales.
 - Diversion Rates must be available on a website and viewable by the general public.
 - Methodology for calculating diversion and recycling rates must be publicly available and applicable to national or country-level accounting standards for construction and demolition waste recycling facilities.
- ▶ Facility recycling data submitted to certification program will be analyzed for recycling rates using a mass balance formula or quarterly sorts completed and verified by an independent third party entity.
- ▶ Final recycling rate will include overall facility diversion rates with and without ADC/Beneficial Reuse, and will include separate recycling rates by material type as well as combined average including wood derived fuel/bio-fuel separate from other waste to energy or incineration end-markets.

Presently, the Recycling Certification Institute's Certification of Real Rates (CORR) protocol meets the above requirements. Find facilities at: www.recyclingcertification.org/certified-facilities

Connection to Ongoing Performance

- ▶ LEED O+M MR prerequisite Waste Performance: A similar measure with modified requirements for existing buildings is required for the O+M v4.1 rating system and is a strategy that can help achieve the MR prerequisite: Waste Performance.